Status of the Magnetized Thermionic Electron Gun

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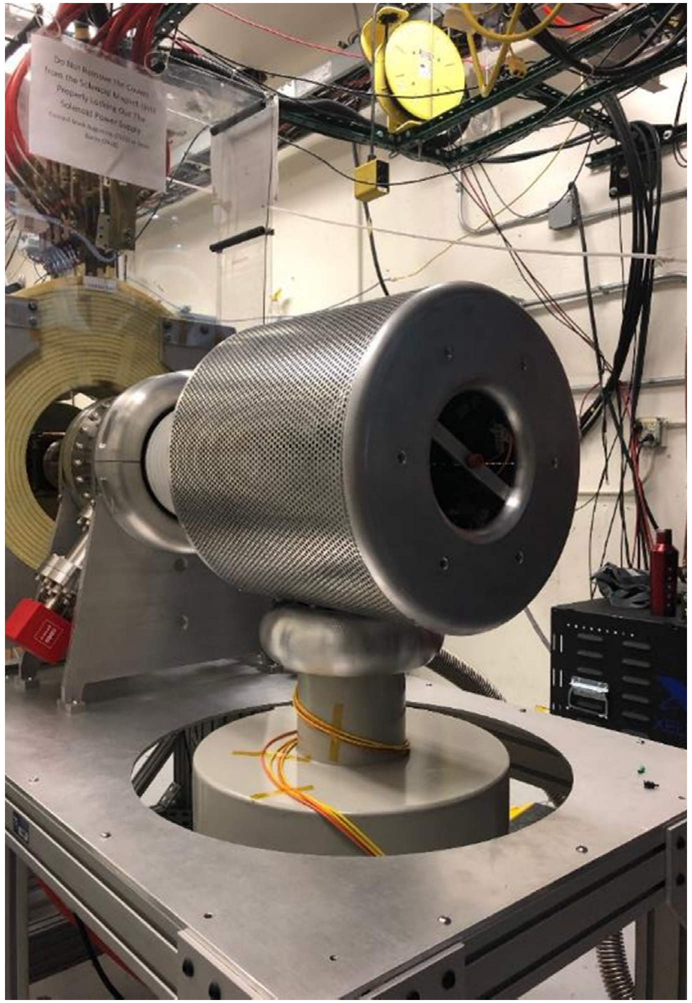

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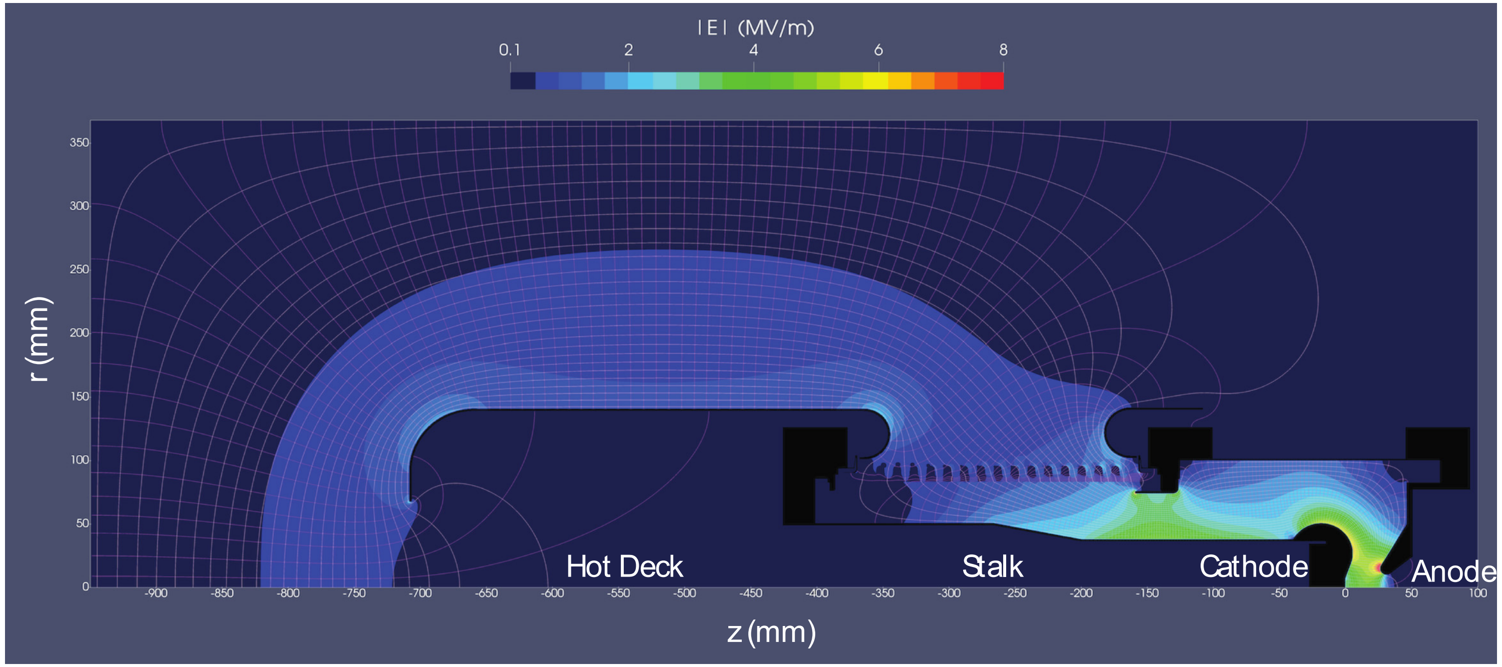
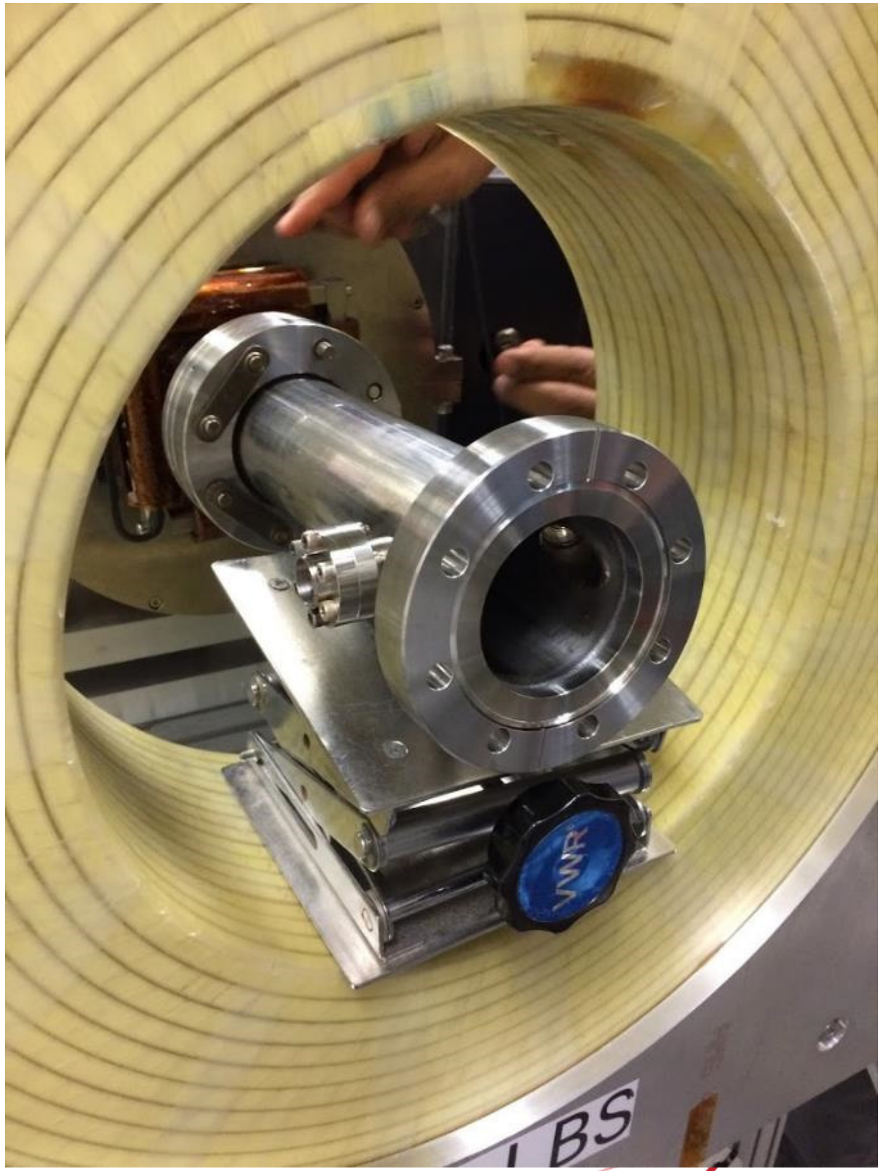
Outline

- ➢ Need for magnetized beams
- ➢ Thermionic gun overview
- ➢ Gun commissioning
- ➢ Diagnostic suite
- ➢ Outlook & summary

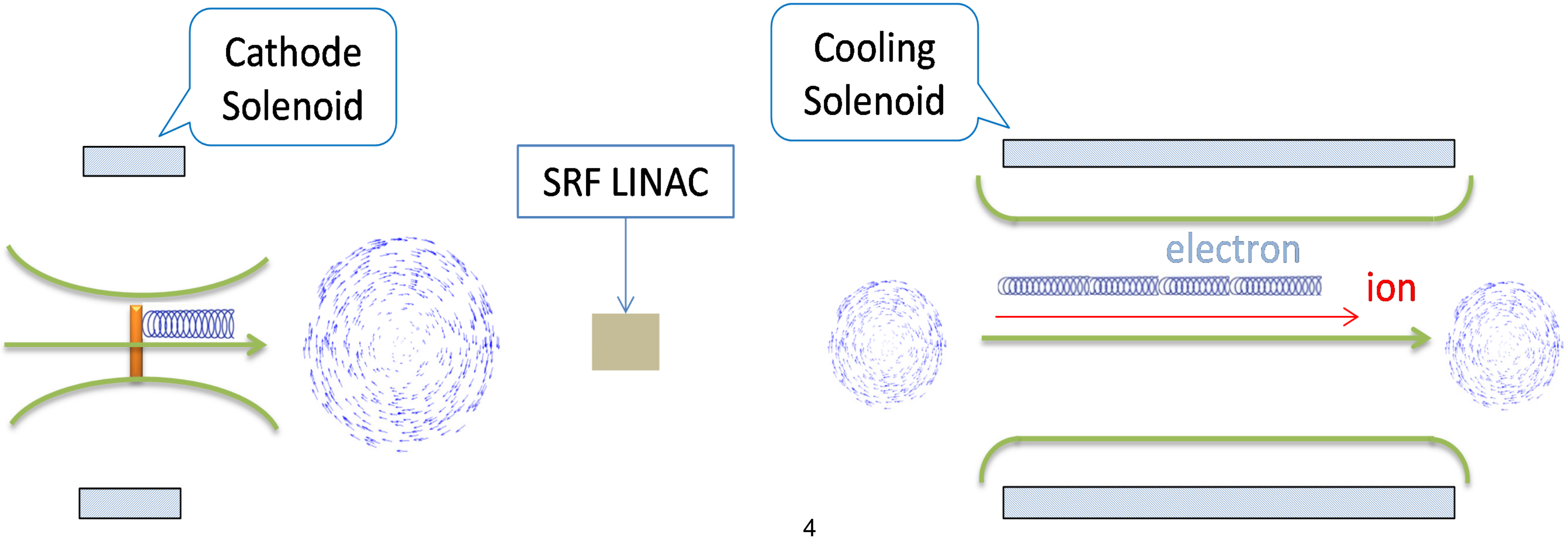




THE NEED FOR MAGNETIZED BEAMS

JLEIC need for magnetized beam

- Electron ion colliders, including JLEIC require electron cooling to improve the luminosity (possibly by a factor of 5).
- Cooling occurs in a long solenoid where electrons and ions co-propagate (30m in JLEIC).
- e-/ion recombination is reduced if the electron beam is magnetized, and not making typical Larmor rotations
- A (pre)magnetized beam becomes ‘quiet’ in the cooling solenoid.
- Magnetized beams are created by immersing the cathode in a solenoid field.



JLEIC Cooler Injector

Current	140mA
Bunch charge	3.2nC
Bunch repetition rate	43.3MHz
Injector Energy	5-7Mev
Uncorrelated transverse emittance	<10µm
Correlated (magnetized) emittance	36µm

- JLEIC invokes an RF photoinjector.
- Photoguns are flexible and offer control over the phase space of the bunch, but have *not been proven reliable at very high currents.*
- A Thermionic gun is being investigated as a viable plan B.
- Through DOE SBIR grant Xelera built a prototype Thermionic Gun (Tgun), and delivered it to JLab's Gun Test Stand (GTS) in July.

Thermionic Electron Source:

- Technology that has been around for 50 years
- Robust, high current source

- Somewhat obsolete with photo cathodes
- Weren't operated in the regime that we require
- Everyone has retired!
- Somewhat re-inventing the wheel

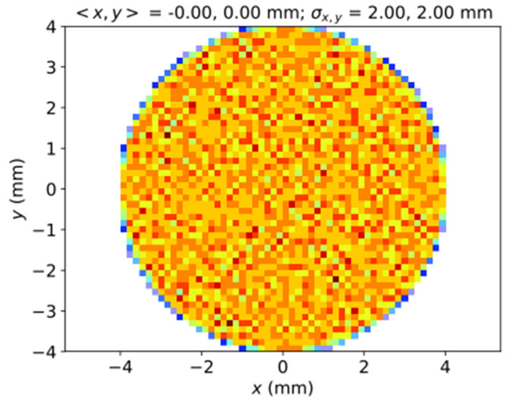
THERMIONIC GUN OVERVIEW

Optimization: Initial Beam/Gun Parameters

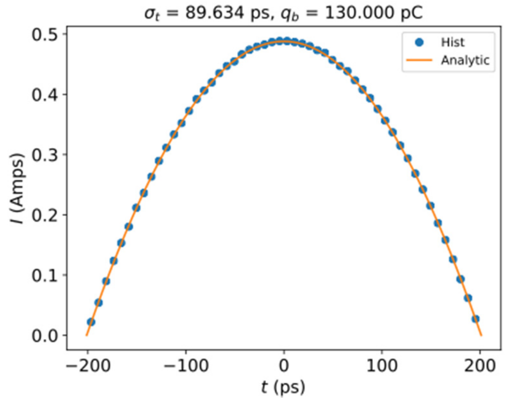
Xelera used MOGO to choose the best electrostatic design for the gun

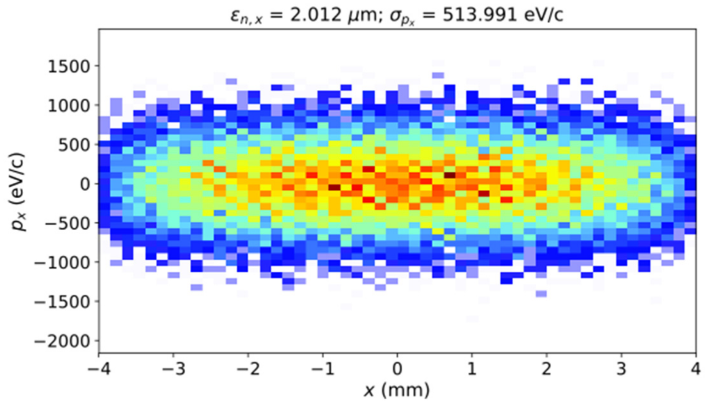
Subsystem	Parameter	Value
Cathode	Emission Radius R_e (mm)	4
	RF Voltage U_{rf} (Volt)	250
	RF Frequency f (MHz)	500
	Transconductance g_{21} (mA/Volt)	10
	Max Temperature T_c (Kelvin)	1500
	Maximum Peak Current $I_{p,\mathrm{max}}$ (A)	1.25
Cathode Solenoid	Cathode Field B_{cathode} (T)	0.0307
Gun	Voltage V (kV)	125
	Cathode-Anode Gap L (mm)	[5, 30]
	Cathode Angle θ_c (deg)	[15, 45]
	Anode Angle θ_a (deg)	[15, 45]
Initial Beam	Bunch Charge q_b (pC)	[0, 545]
	RMS Bunch Length σ_t (ps)	[0, 147]
	Half Emission Angle ψ (deg)	[0, 60]
	Thermal Emittance $\epsilon_{n,\mathrm{therm}}$ (μm)	1.0
	Initial Emittance $\epsilon_{n,x,0} = 2\epsilon_{n,\mathrm{therm}}$ (μm)	2.0
	Magnetized Emittance $\epsilon_{n,\mathrm{mag}}$ (μm)	36
	Canonical Angular Momentum $\langle \mathcal{P}_\theta \rangle$ (neV·s)	122.8
	Transverse Thermal Energy $k_B T$ (meV)	129.26
	Effective Mean Transverse Energy $\mathrm{MTE}_{\mathrm{eff}}$ (meV)	517.04

Transverse distribution @ cathode

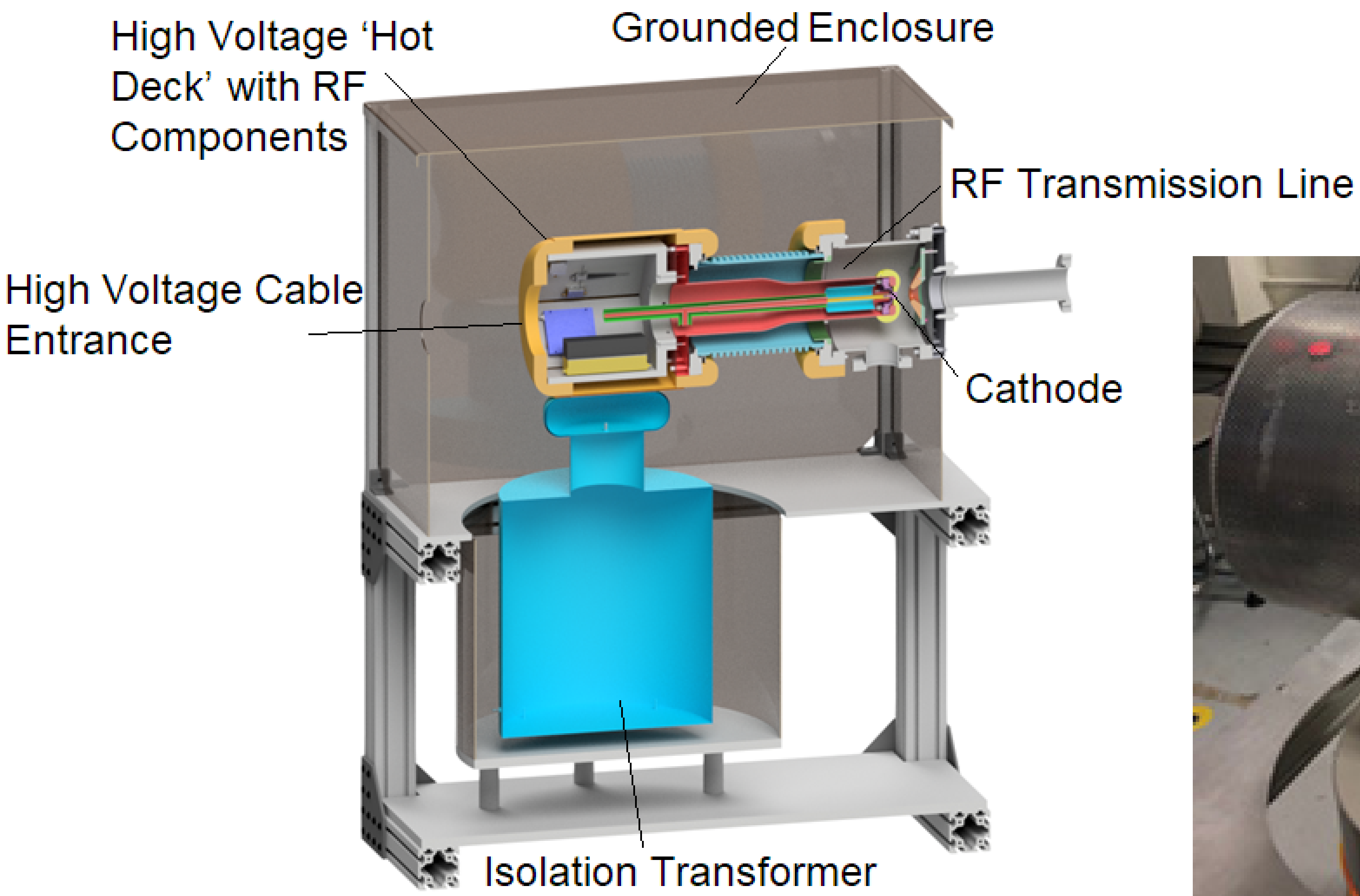


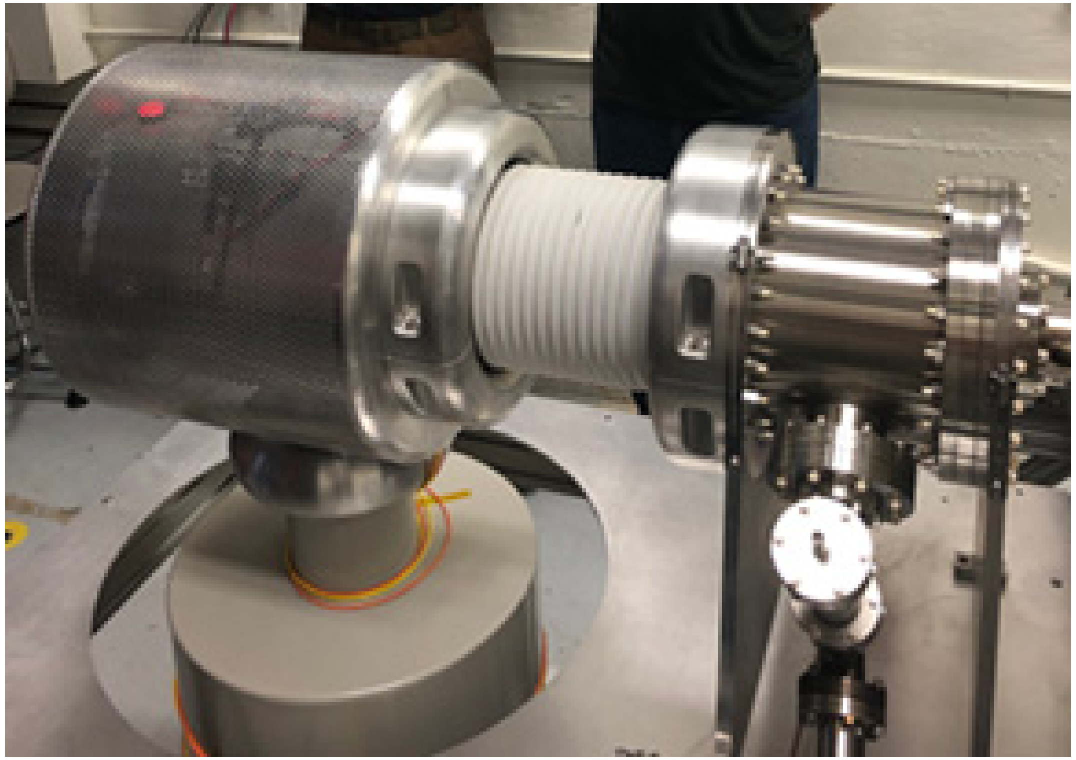
Longitudinal distribution @ cathode



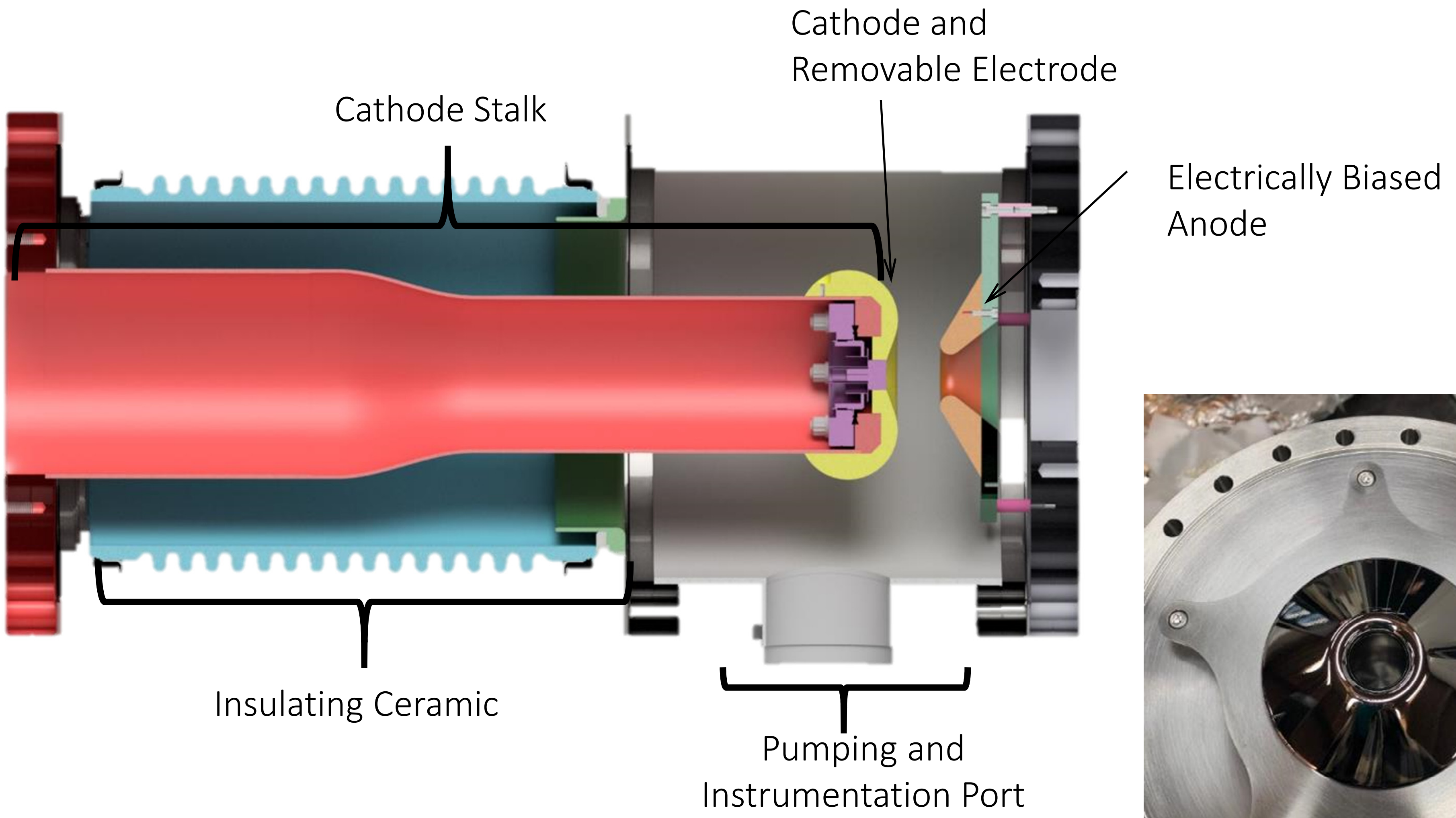


Mechanical Design

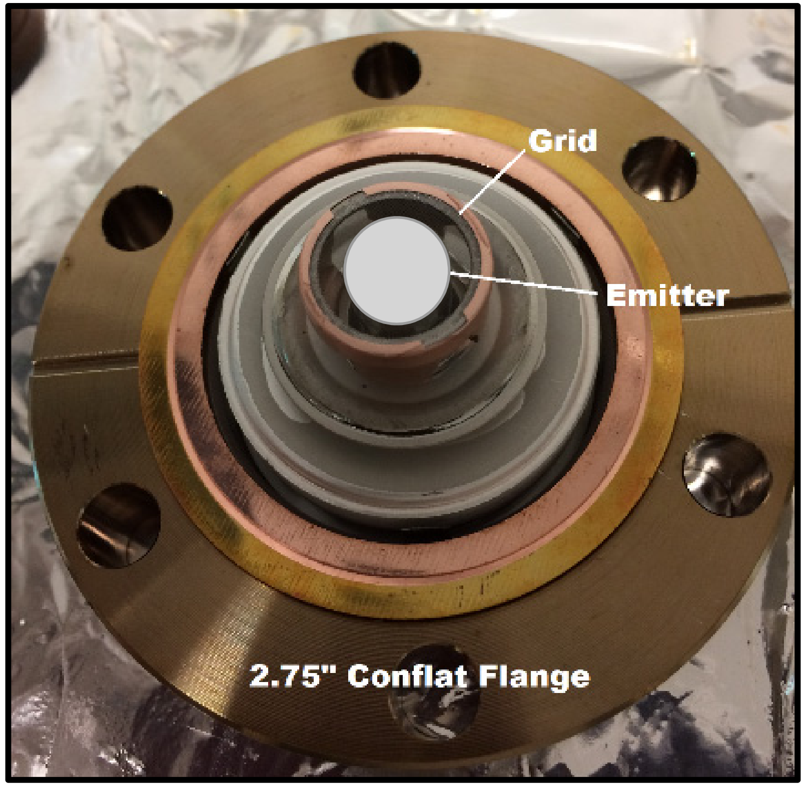




Vacuum System Mechanical Design

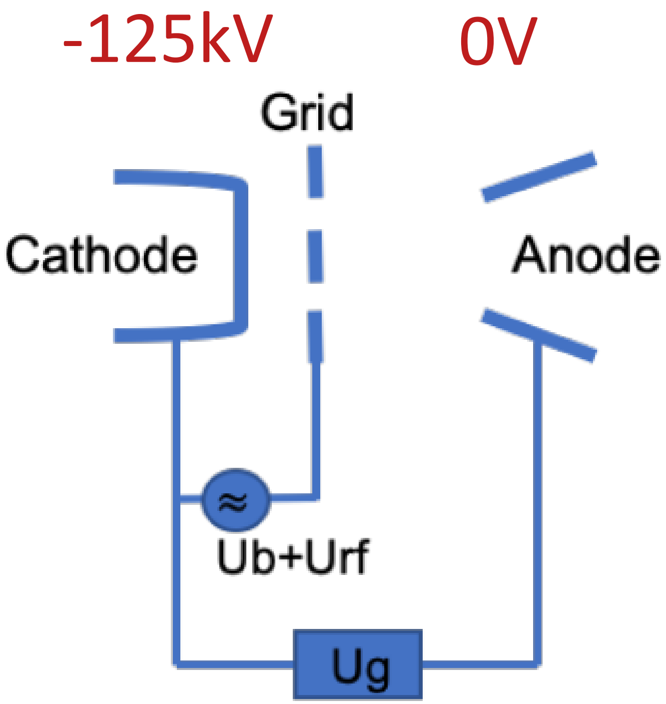


Cathode – commercial off-the-shelf

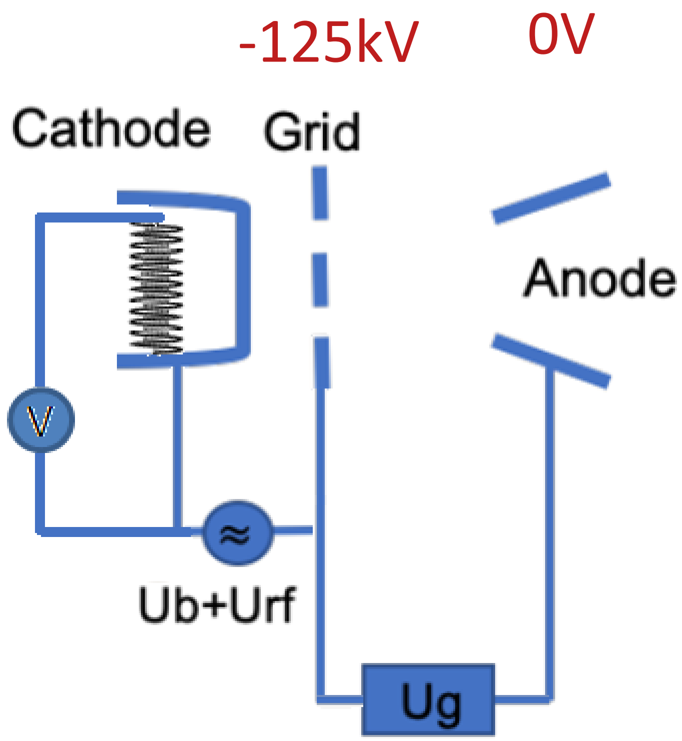


Vacuum side

Reverse



Cathode connection



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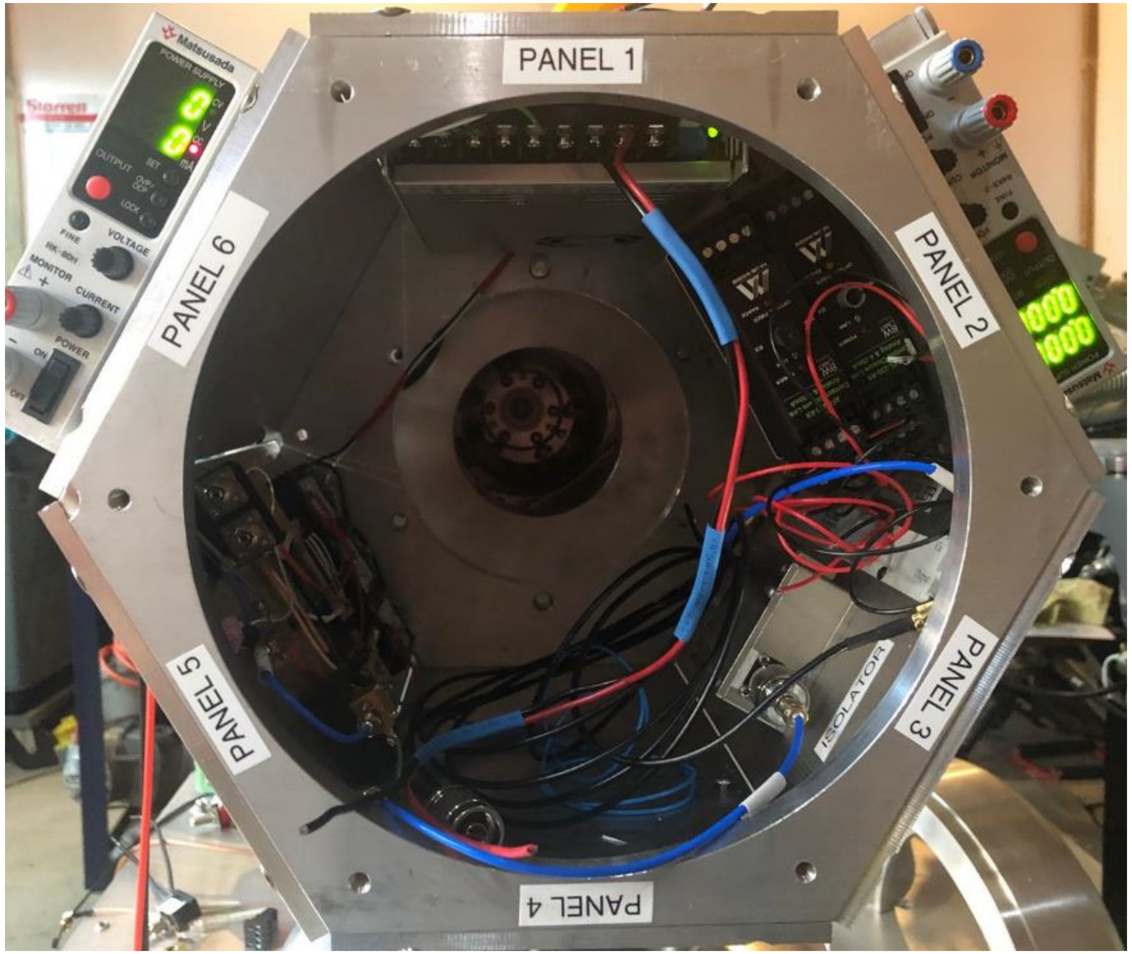
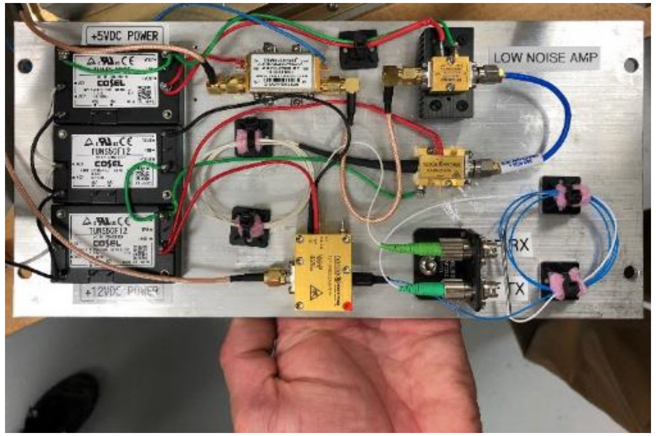
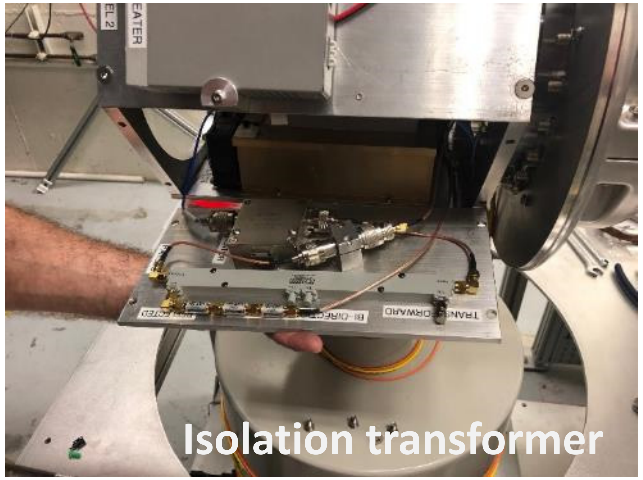
Hot Deck Electronics, RF Feed with tuner

Hot deck



Internal panels





RF transmission line

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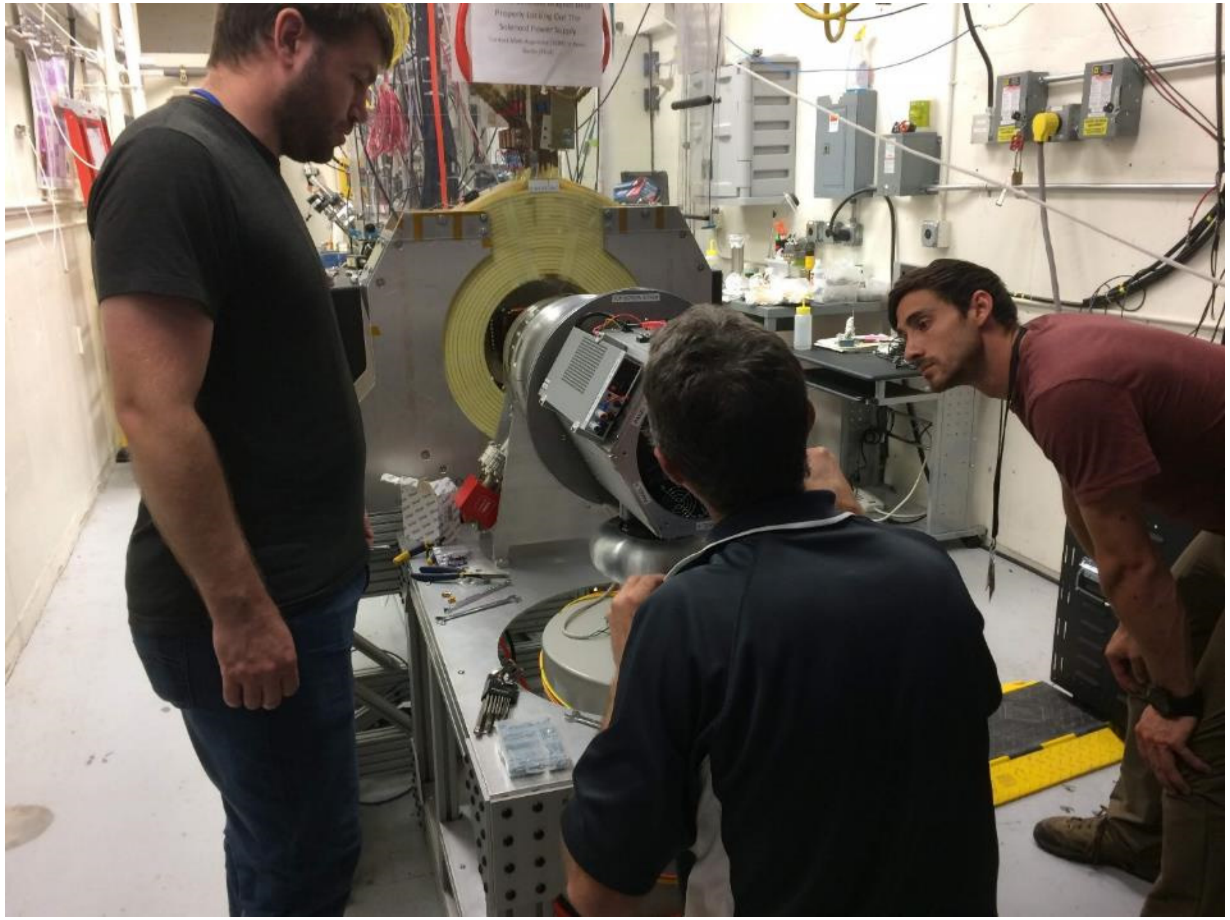
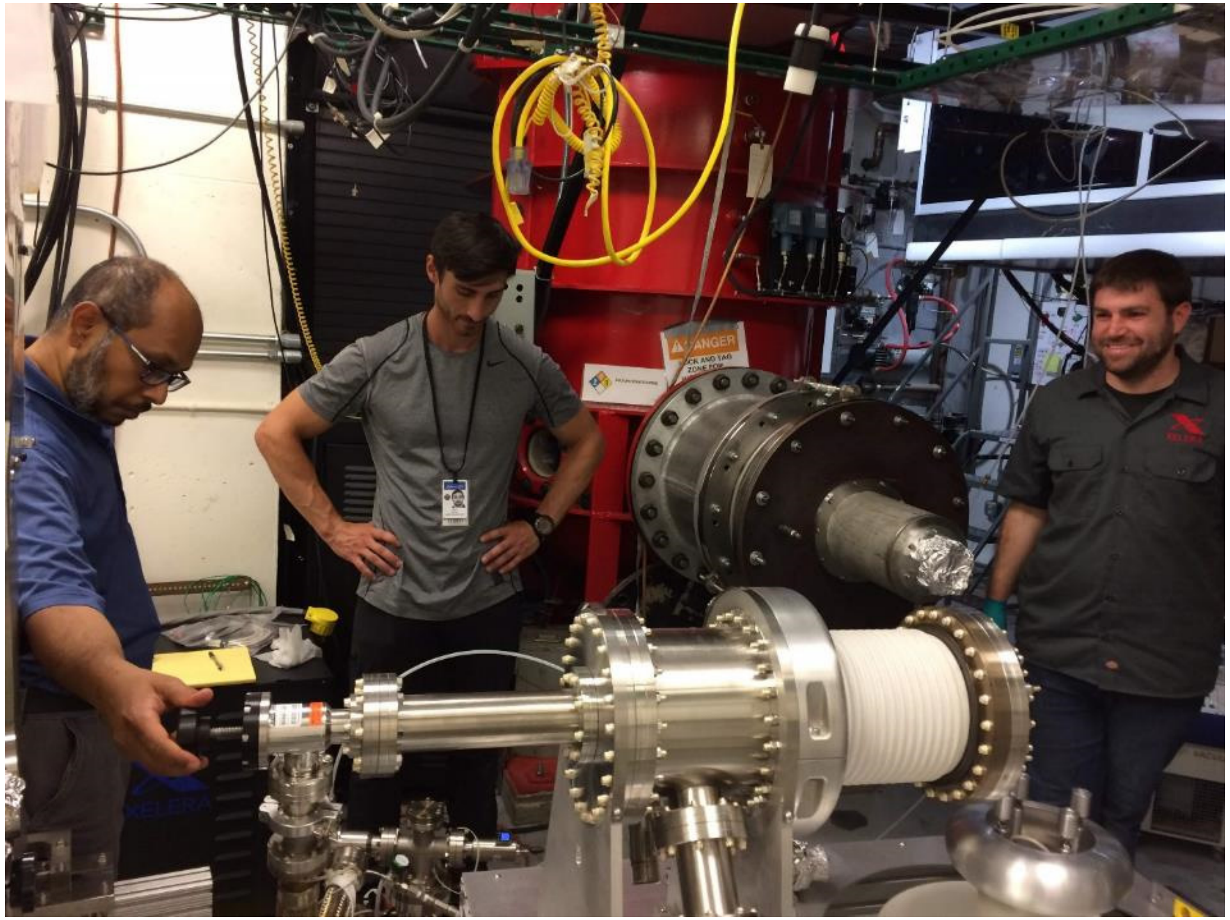
THERMIONIC GUN COMMISSIONING

Testing in JLab’s Gun Test Stand





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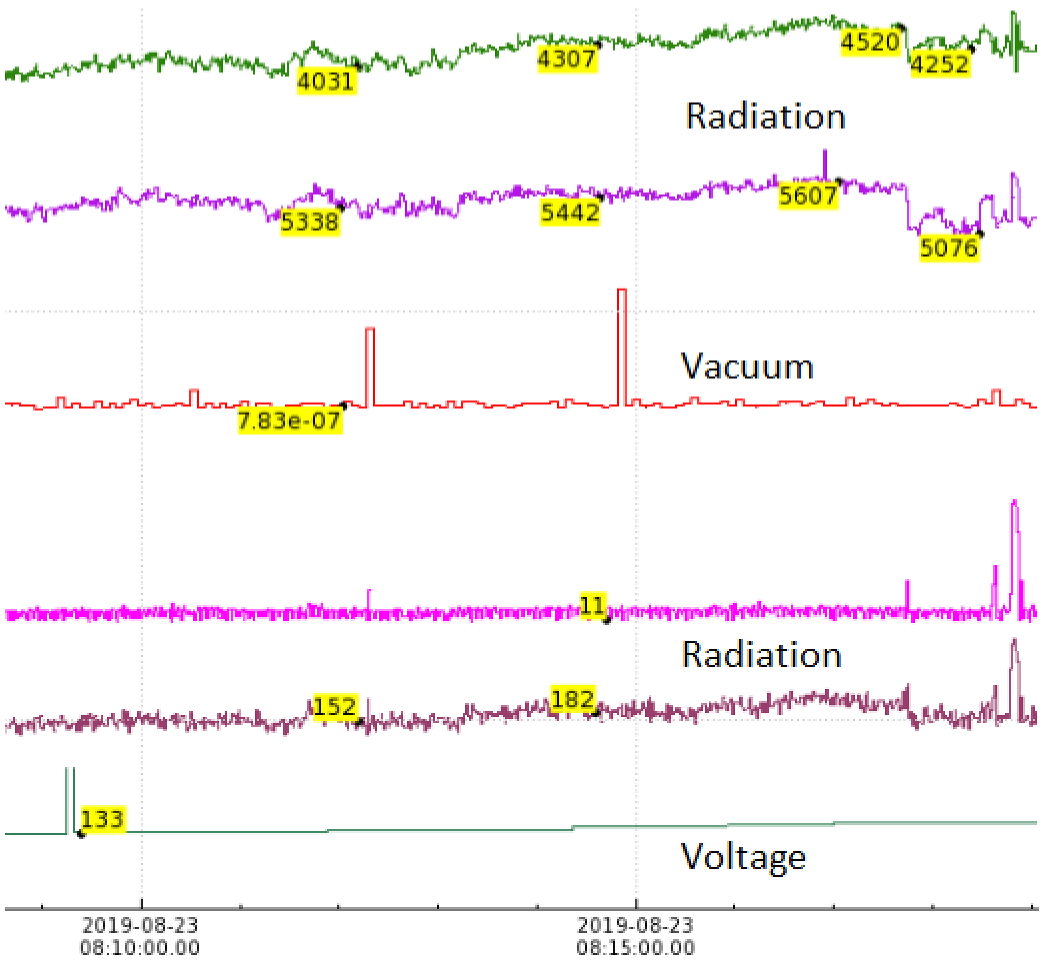
Vacuum preparation

- Bake took 4 days
- 3e-10 Torr before bake (large vacuum bursts with HV conditioning-water)
- 3.8e-10 Torr after 150C bake (vacuum burst brief)
- 1e-10 Torr when idle
  - 5e-9 Torr with HV on.

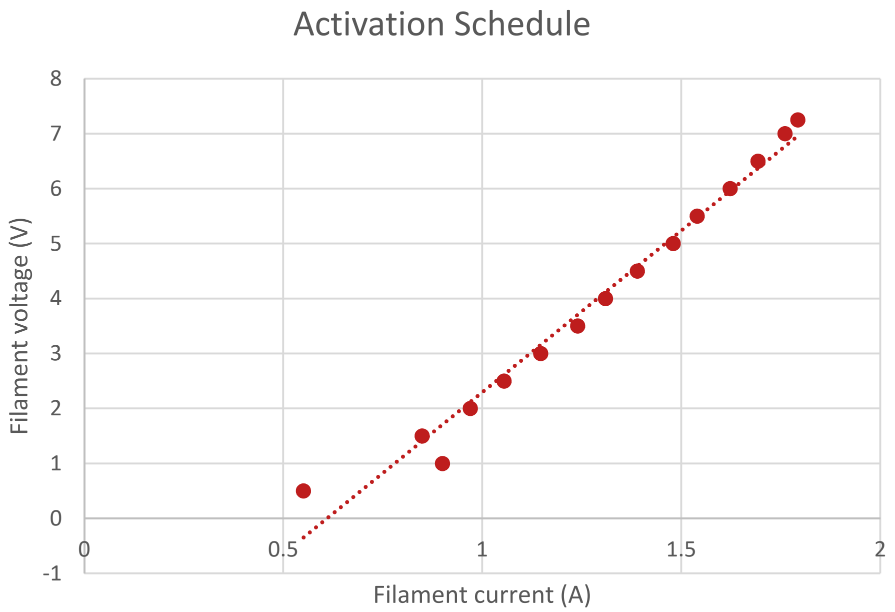
Initial testing – the gun

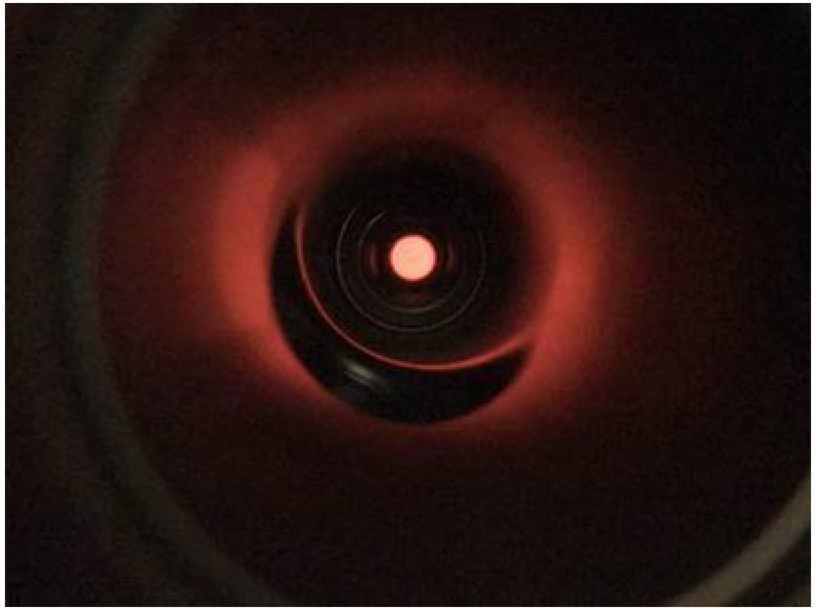
- High Voltage Processing
  - 140kV (0.3mA Field emission current, onset 90kV)
  - Experiencing some HV power supply issues (EMI/EMC tripping and interfering with nearby electronics) – (Please come talk to me)
- Kr processing
  - Has been successful with field emission suppression in photoguns
- Gun has been at 120kV for 2 days without tripping. Vacuum in e-8 Torr range.



Initial Testing – the cathode

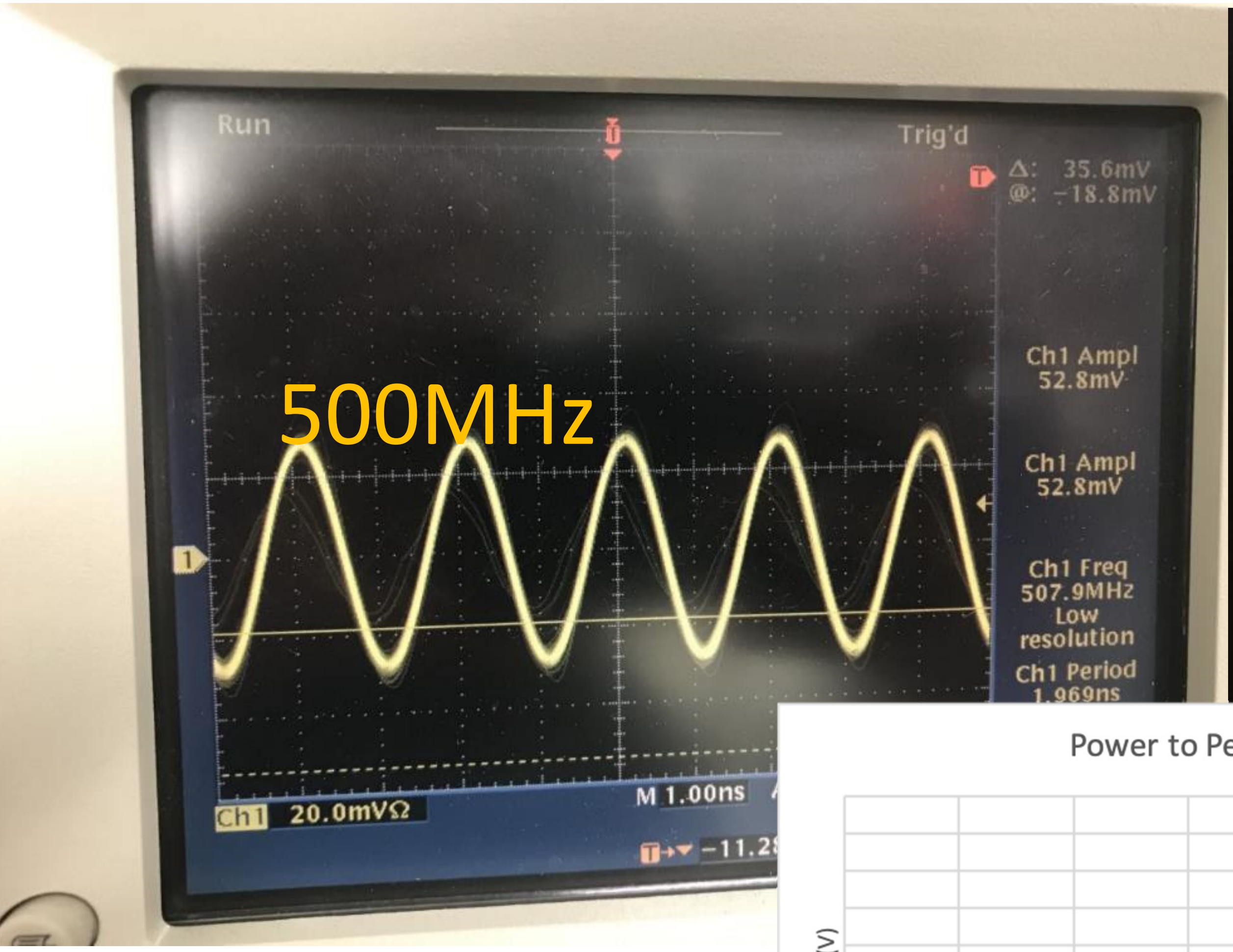
- Cathode activation
  - Increase filament voltage/current whilst maintain low vacuum
- Worked fine until the bake
  - Odd grounding of the grid, couldn’t pull current

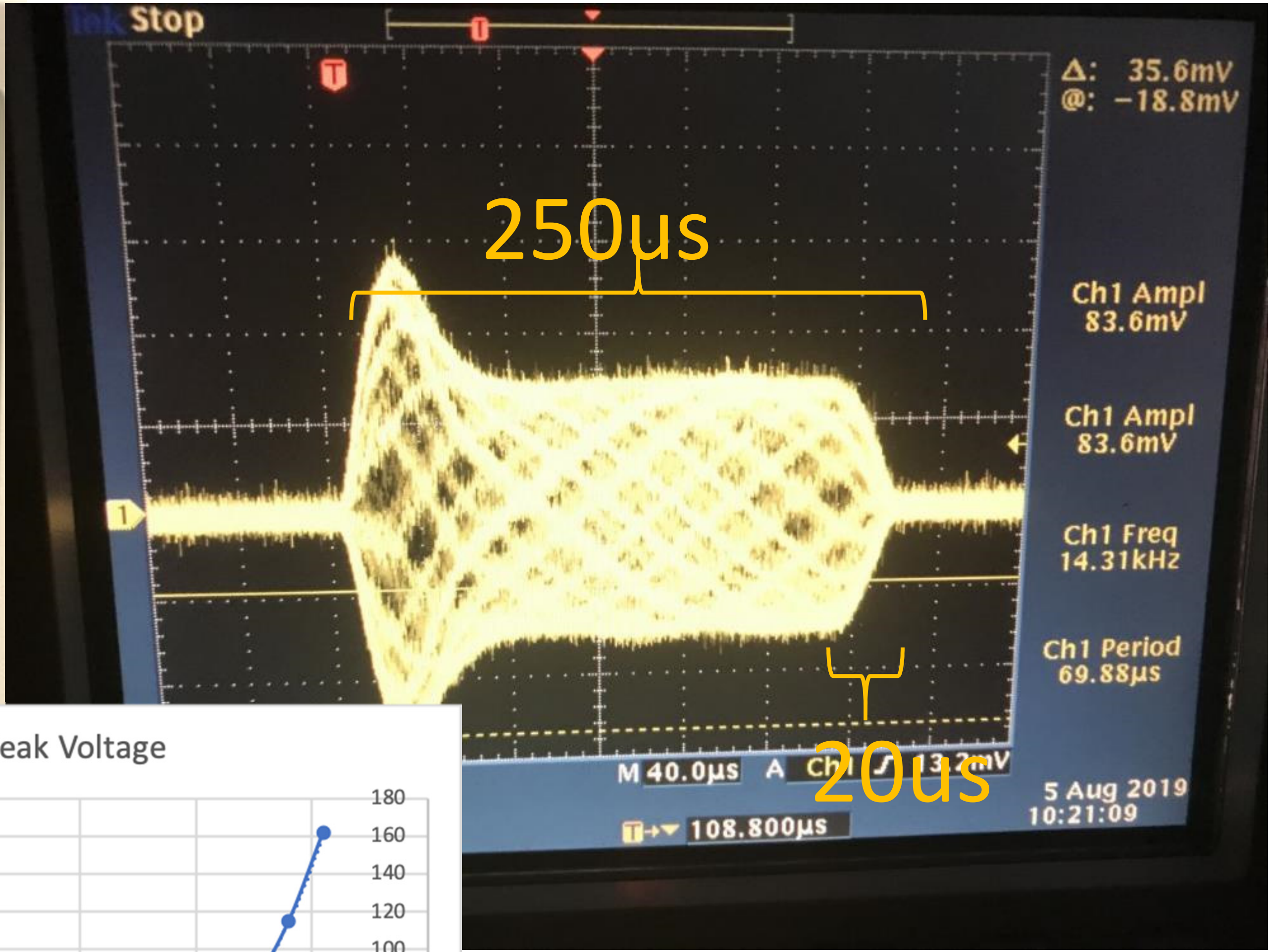


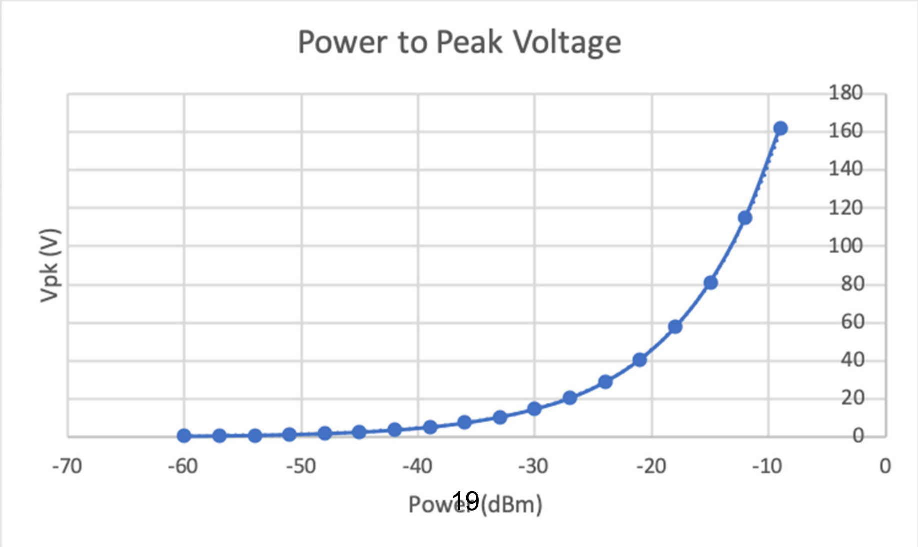


Warm cathode, ~3V

RF testing







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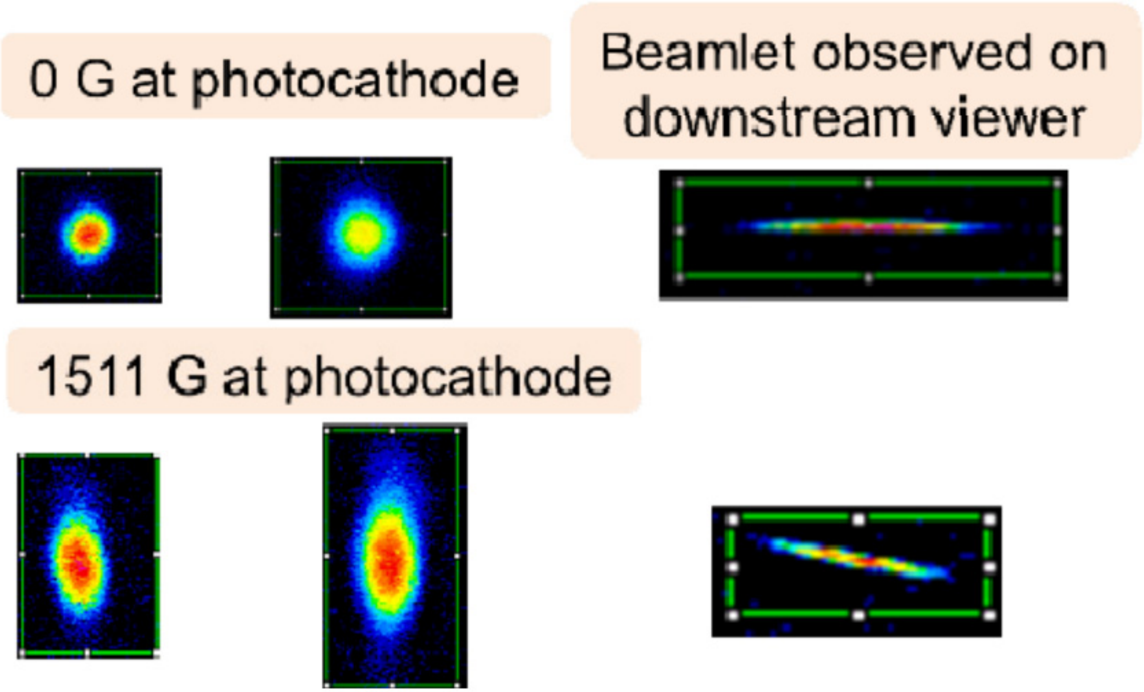
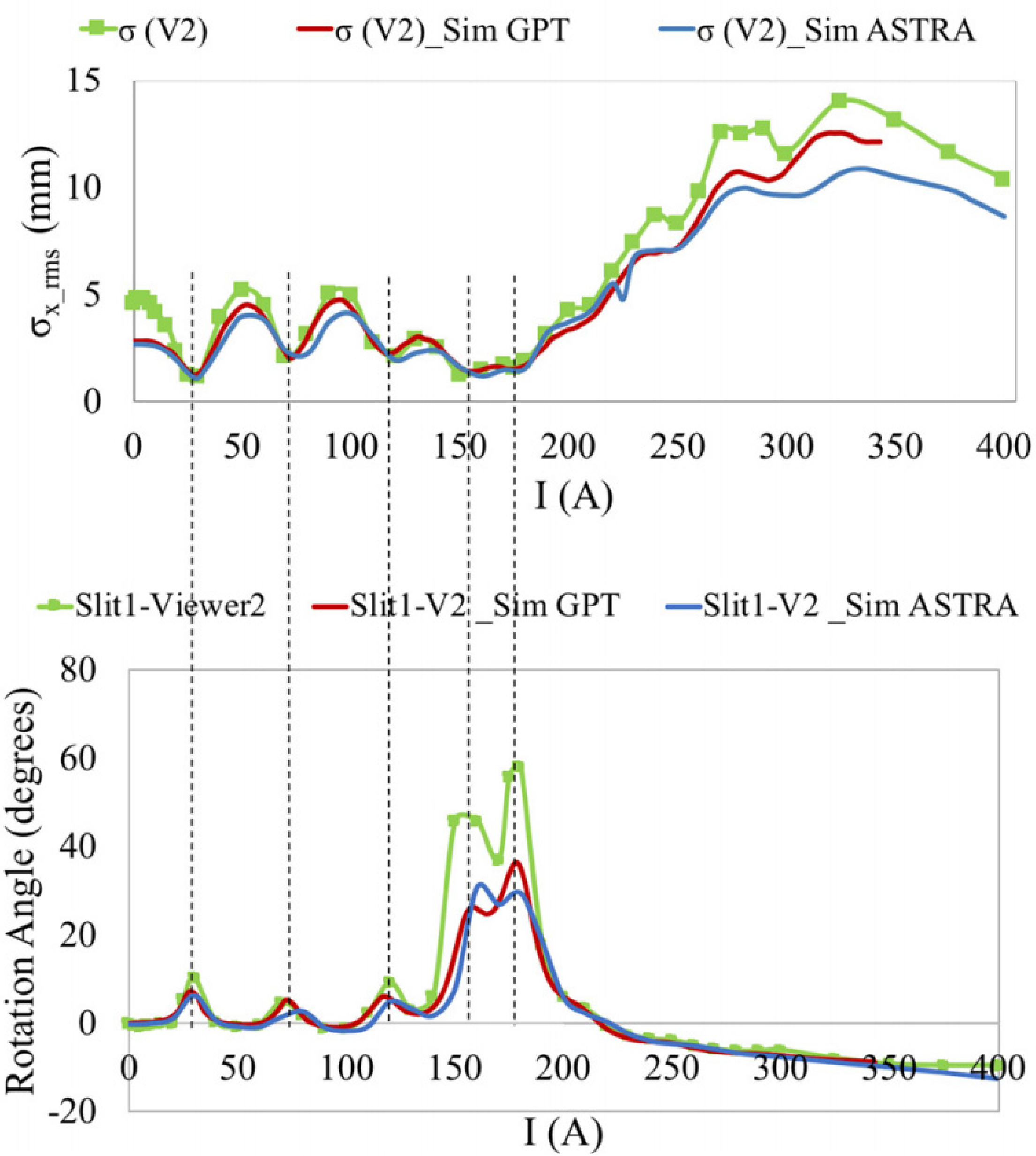
PROPOSED MEASUREMENTS

E- beam measurements: Magnetization Effects

- Beam size and rotation vs B field @cathode

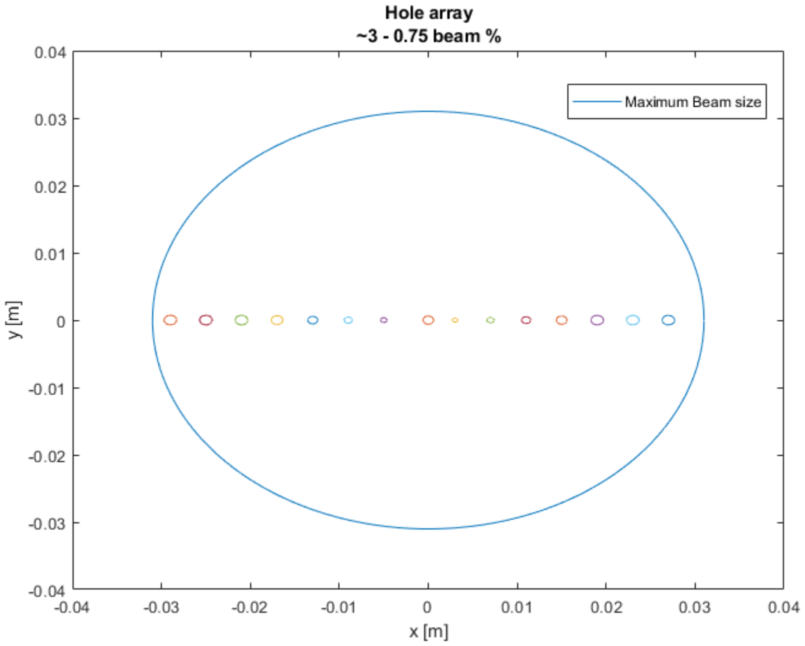
Examples from our magnetized photogun

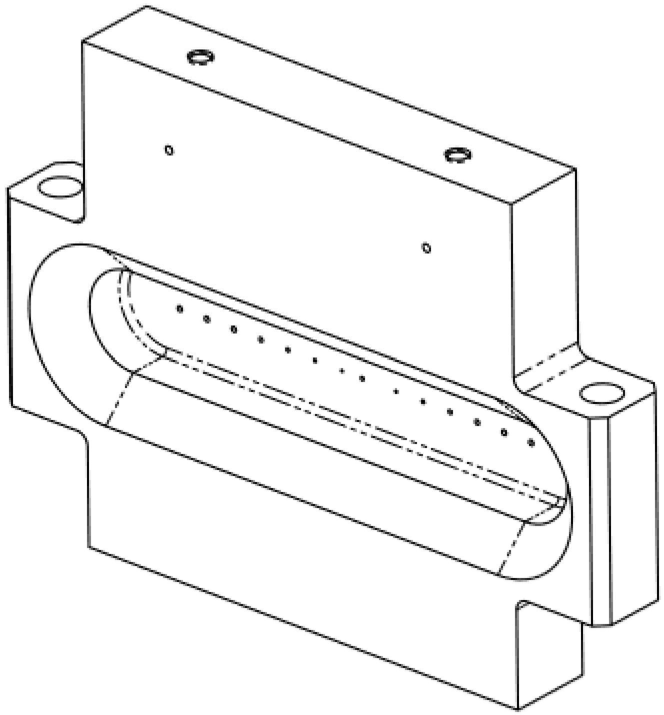


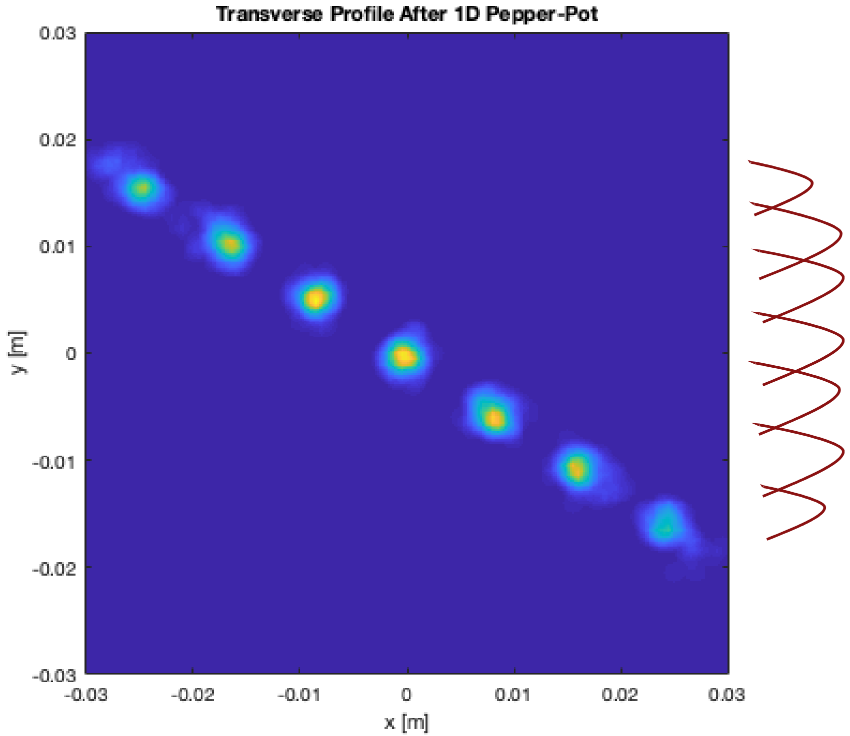


E- beam measurements: Transverse Emittance

- 1D Pepper-pot (tungsten)
- Have performed virtual experiments

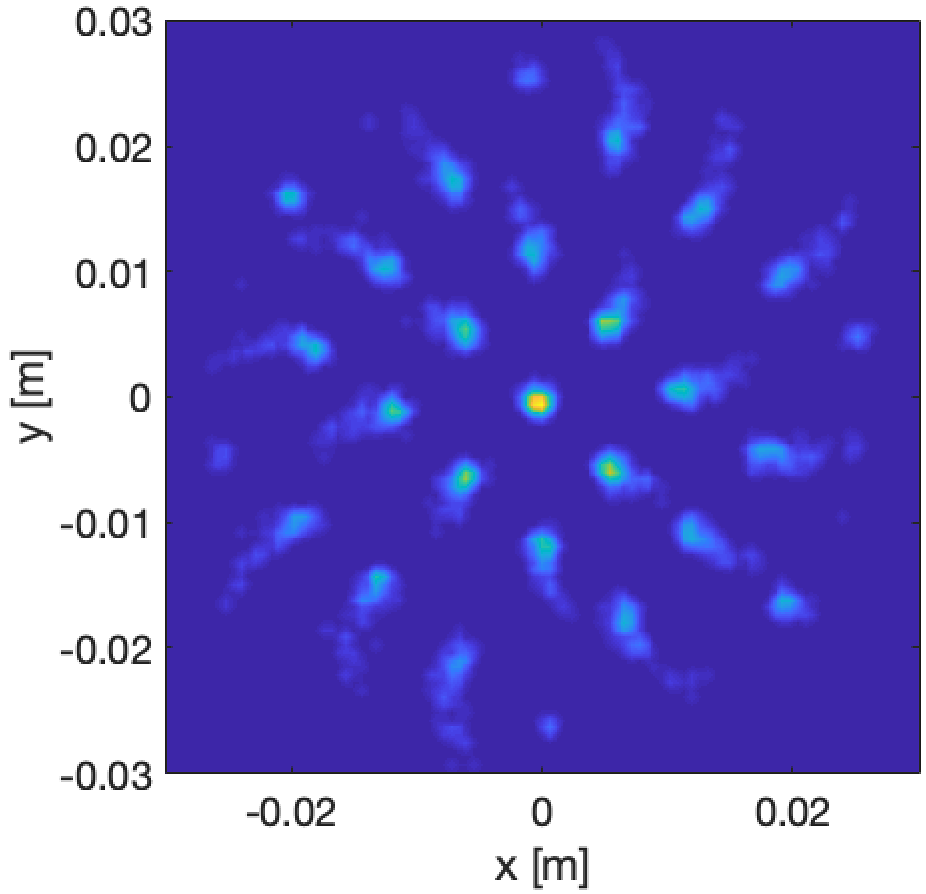




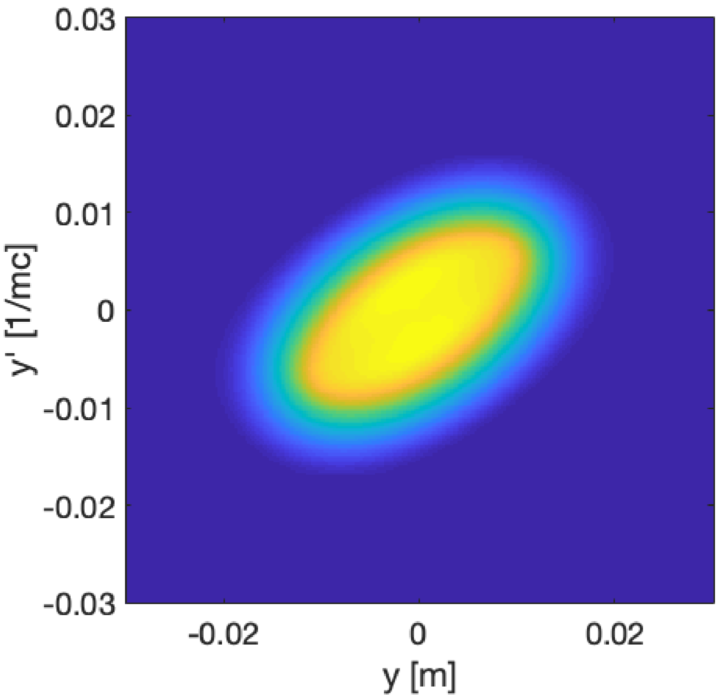


E- beam measurements: Transverse Emittance

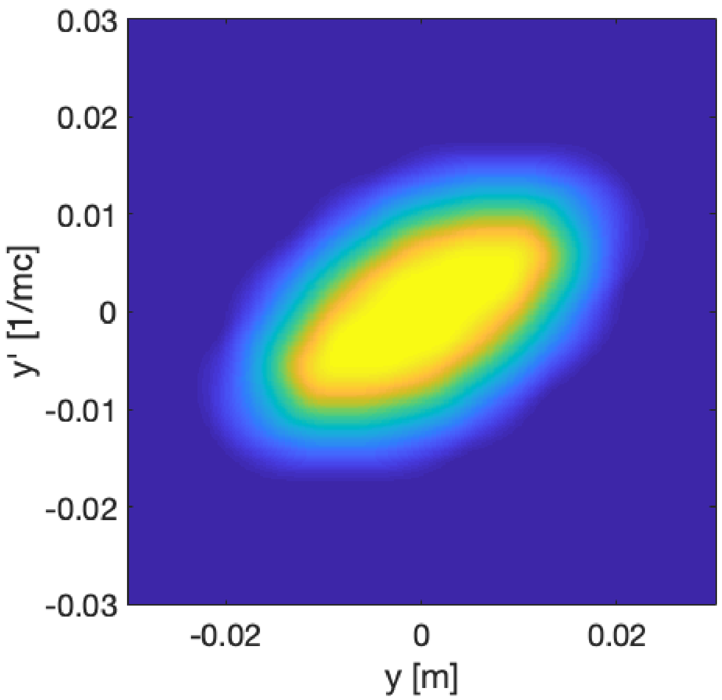
- Can scan vertically and horizontally to get both transverse planes
- Post process in the typical way to reconstruct phase space



Superposition of 5 scans



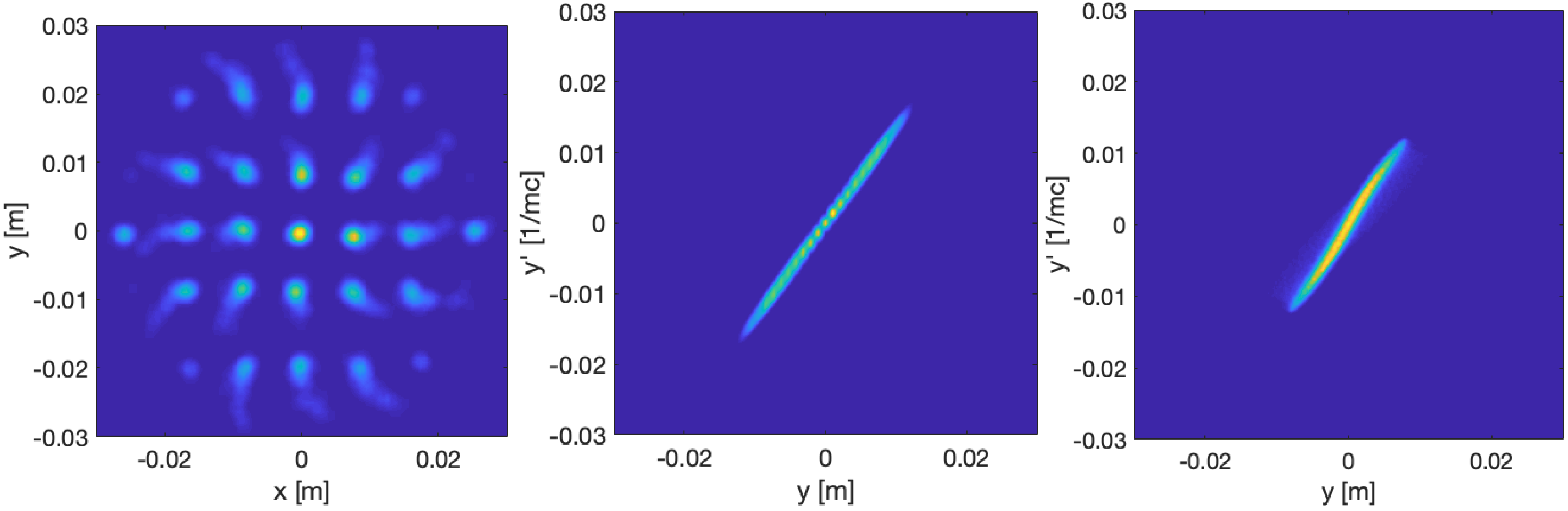
Reconstruction with 19 slices



Directly from simulation

E- beam measurements: Transverse Emittance

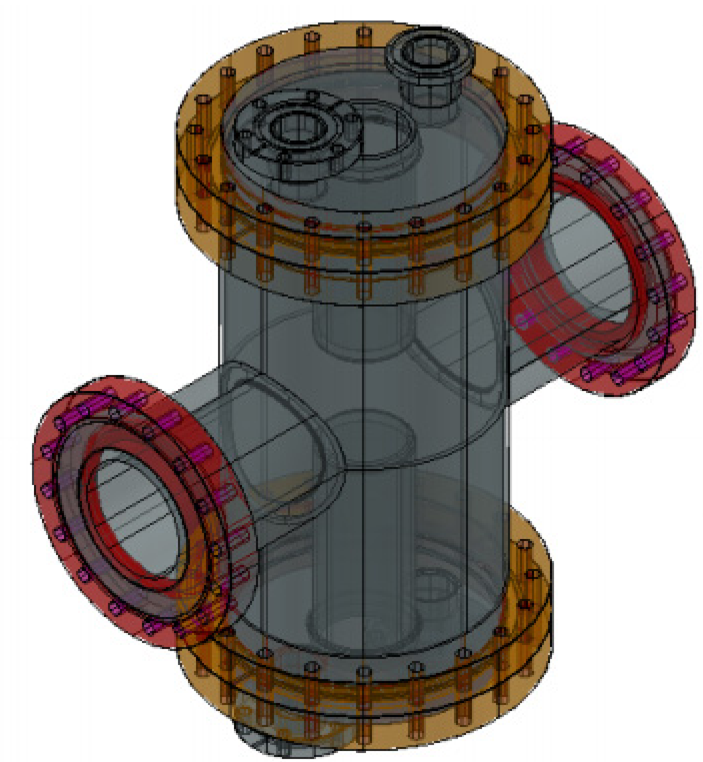
Can also remove the angular momentum rotation to recover the uncorrelated transverse emittance (non-magnetized portion)

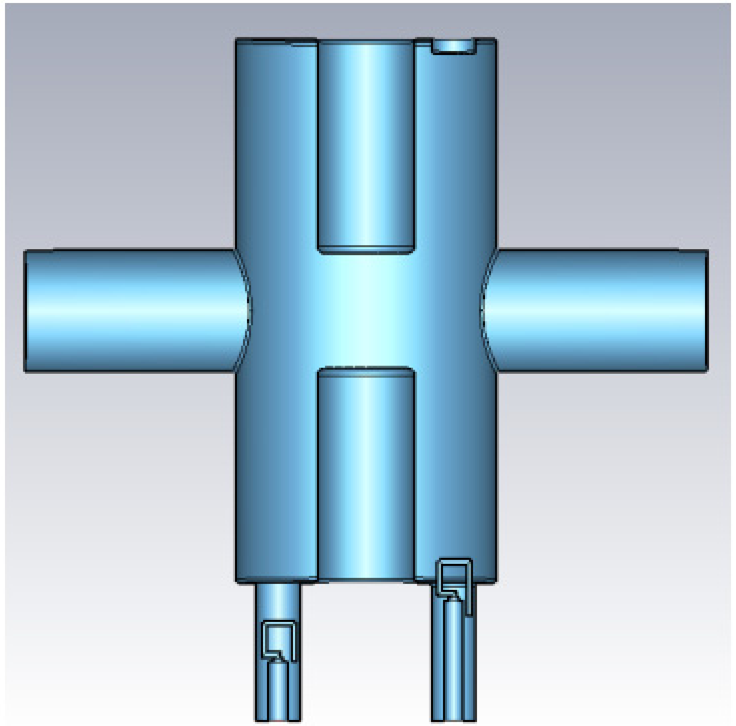


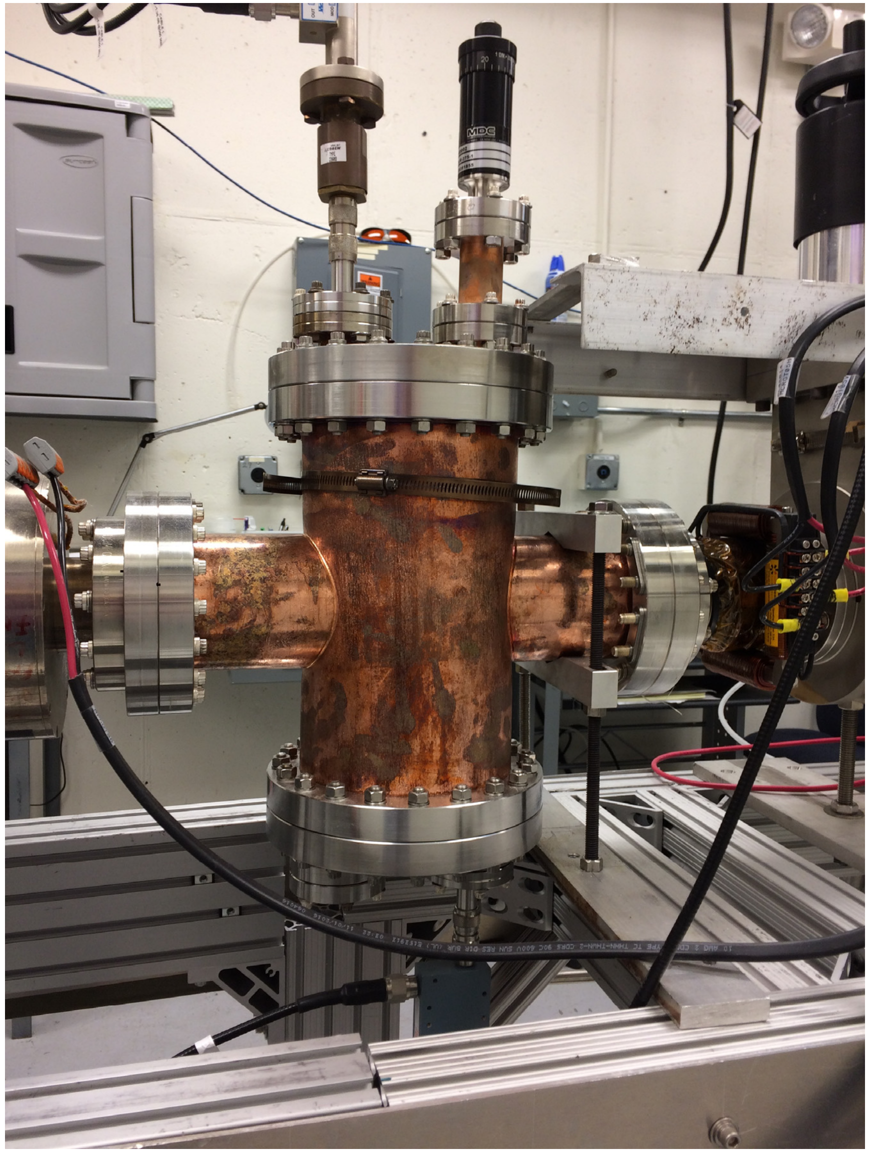
Un-rotated | Reconstructed from 19 slices | Directly from simulation

E- beam measurements: Longitudinal profile

- Built a TE double quarter wave deflecting cavity
- Want to measure the sinusoidal longitudinal profile
  - Determine how well RF is applied to the emitter/grid







Mark Stefani

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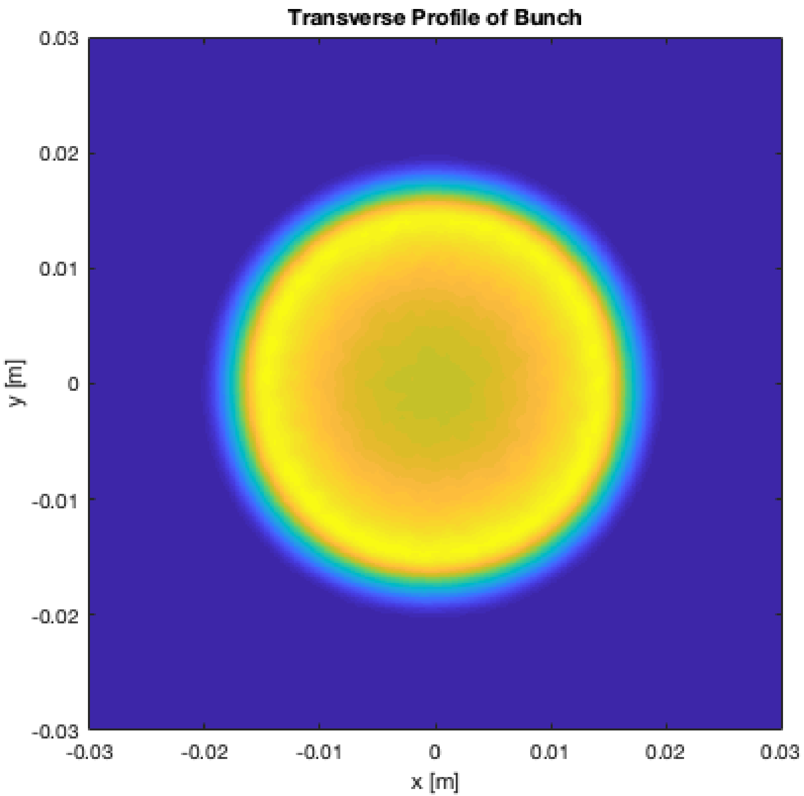
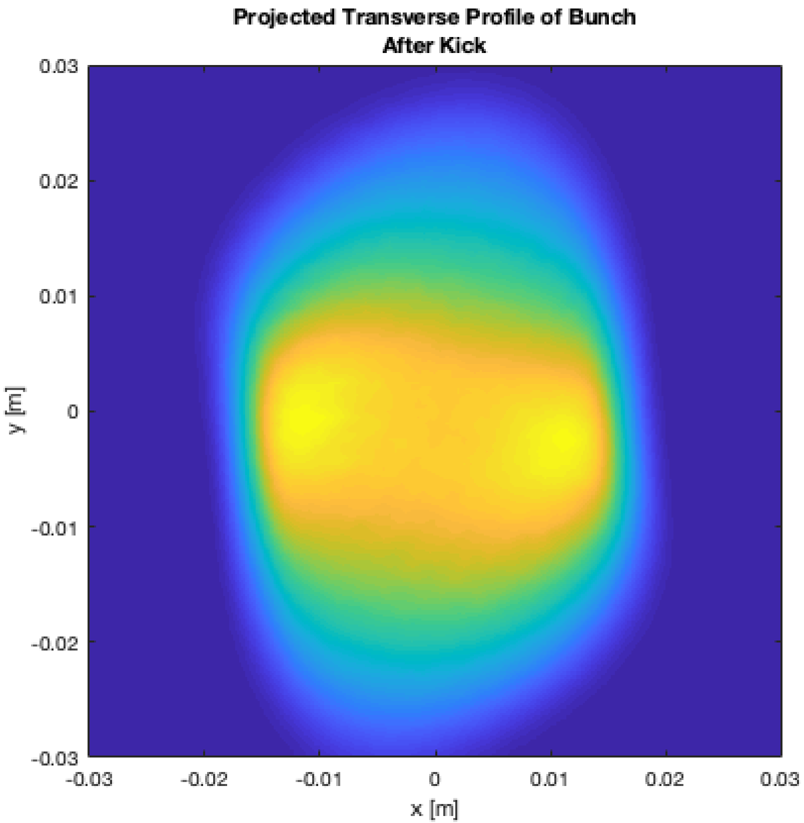
E- beam measurements: Longitudinal profile

- Deflecting the full beam (which is large transversely and tilted) is difficult to analyze



3cm FWHM typical beam



Not much deflection to fill viewer screen
- Not at all accurate

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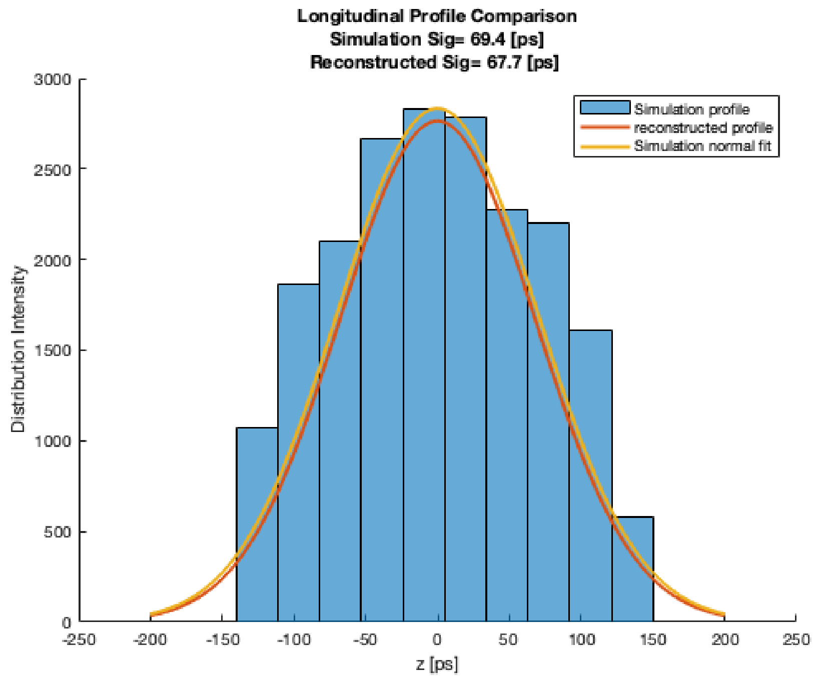
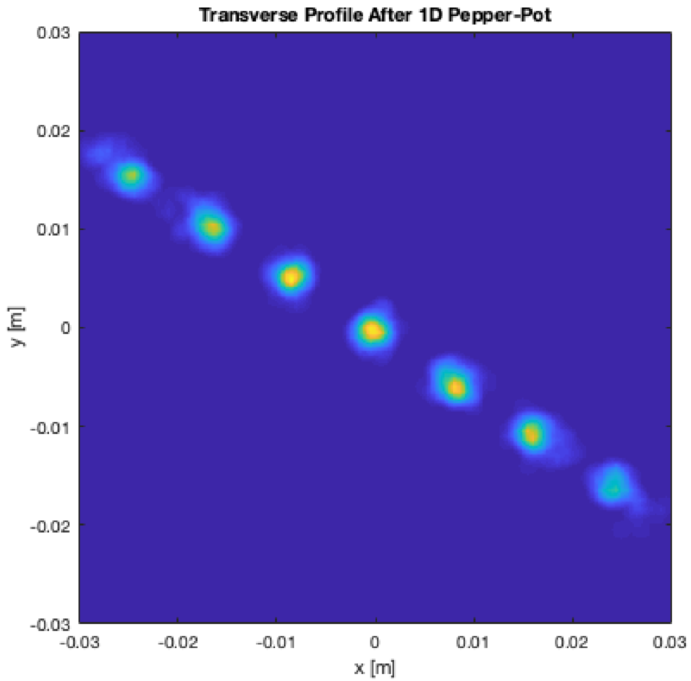
E- beam measurements: Longitudinal profile

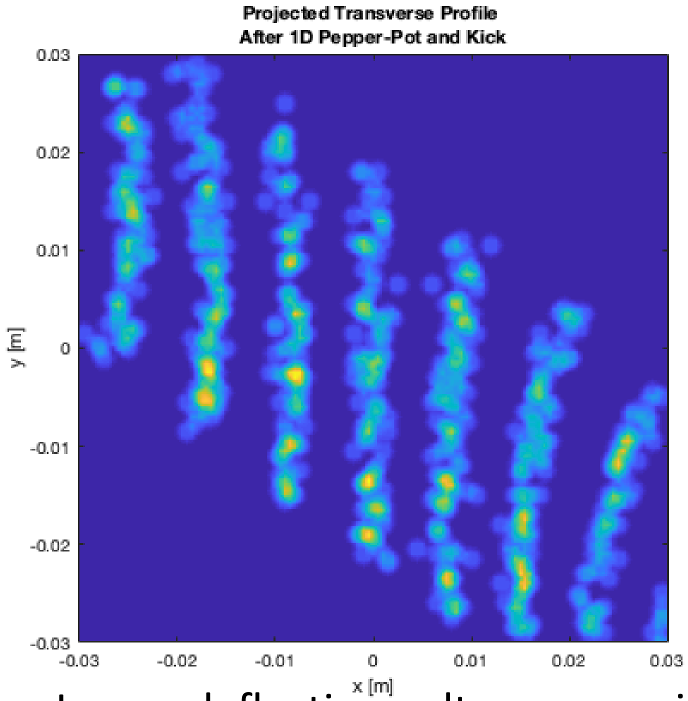
- Again use the 1D pepper pot to aid the results
  - Gives profile as a function of horizontal position
  - Shows the magnetization rotation too
  - Remove the rotation to yield the profile



Virtual experiment and simulation show good agreement.



Sample core of beam with 1DPP



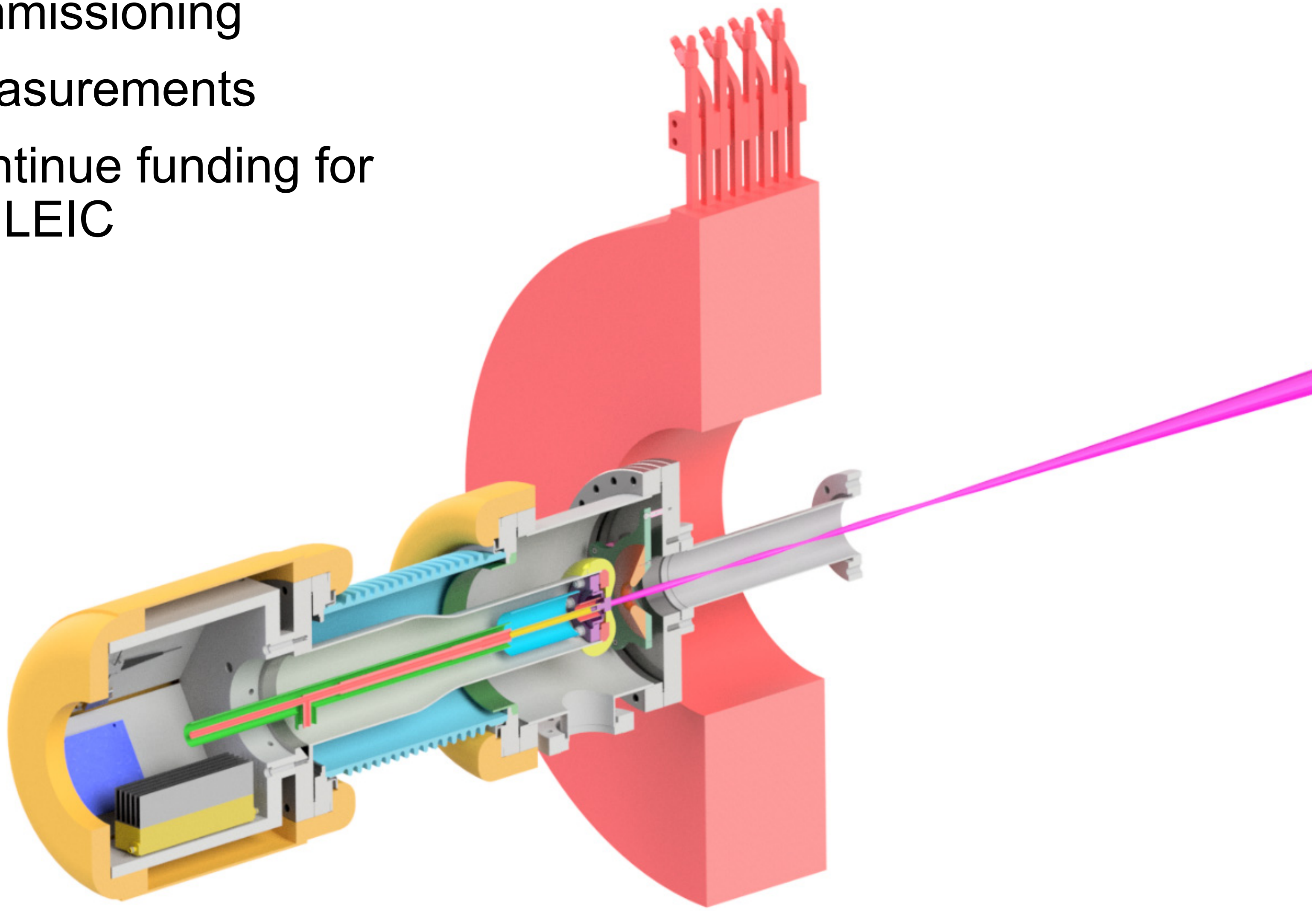
Larger deflection voltage on cavity (more accurate)

CONCLUSION

Outlook

- Making steady progress with commissioning
- Poised to take electron beam measurements
- Will collaborate with Xelera to continue funding for a higher voltage gun that meets JLEIC specification.

- Thank you!



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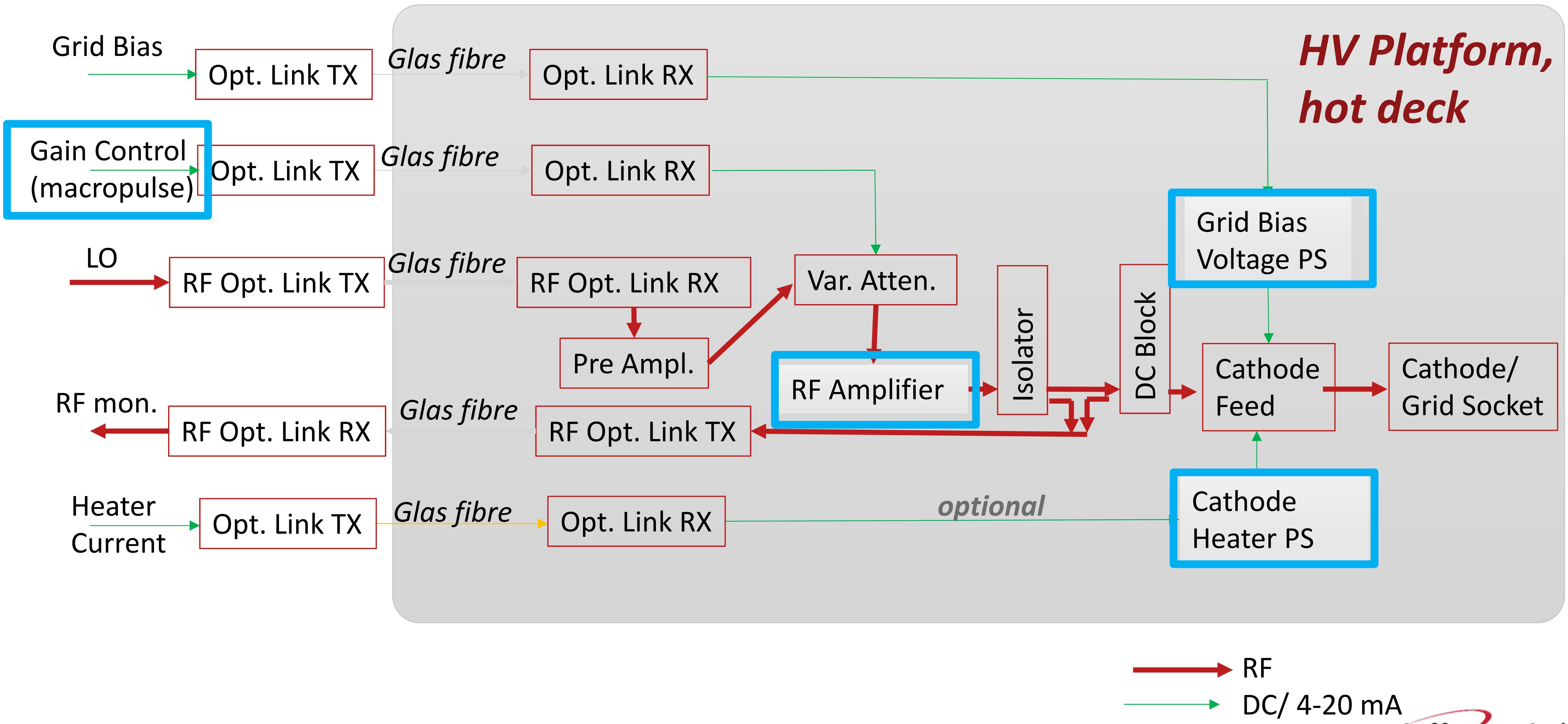
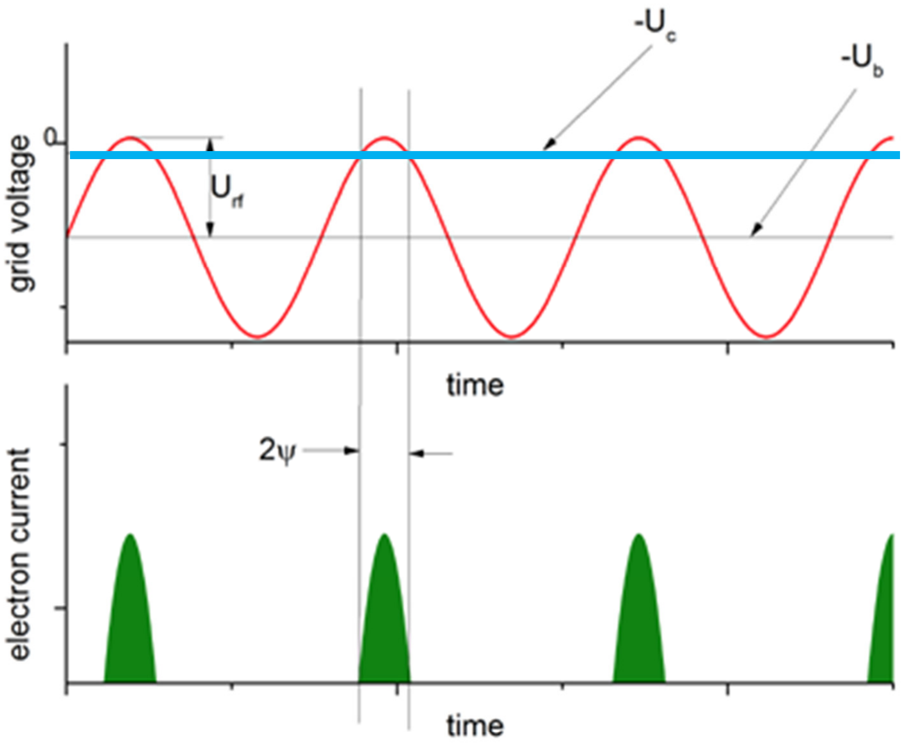
RF/ Cathode & Grid Drive Design



Cathode emission model



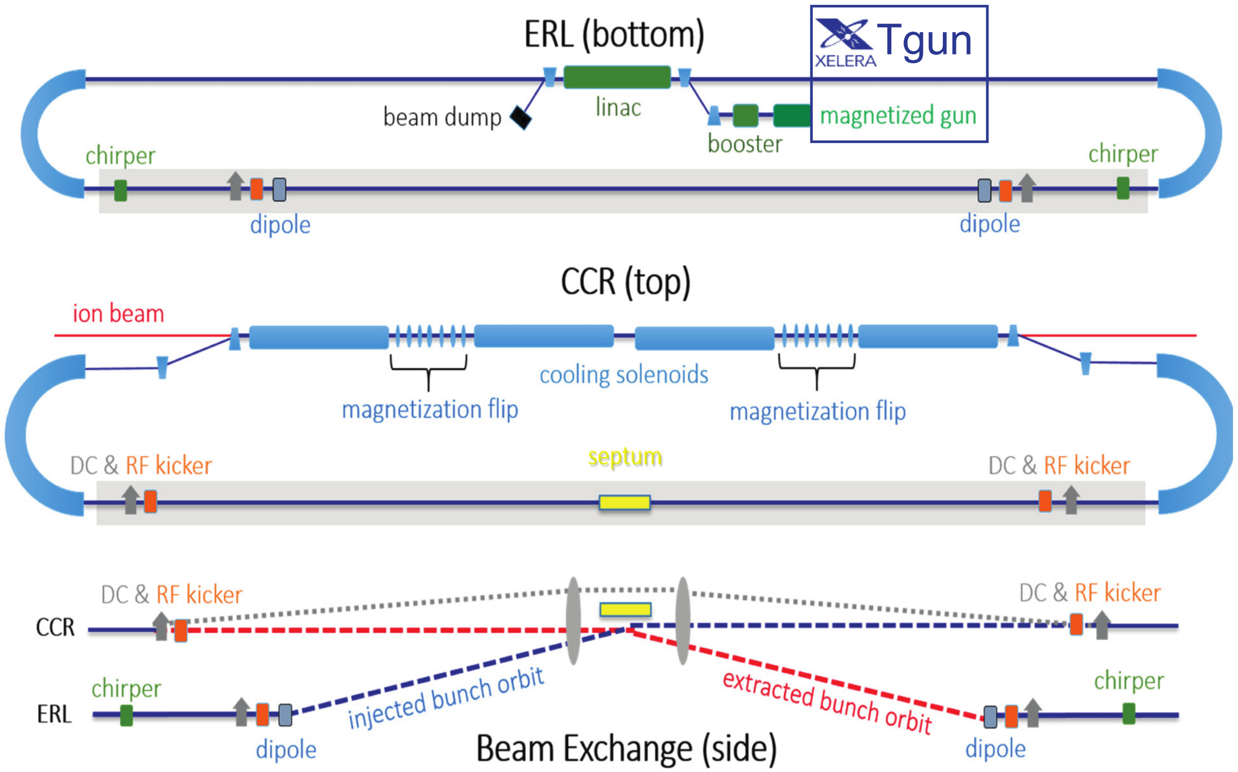
Cathode transconductance

$$q_b = \frac{2g_{21}U_{\mathrm{rf}}}{\omega}(\sin\psi - \psi\cos\psi)$$

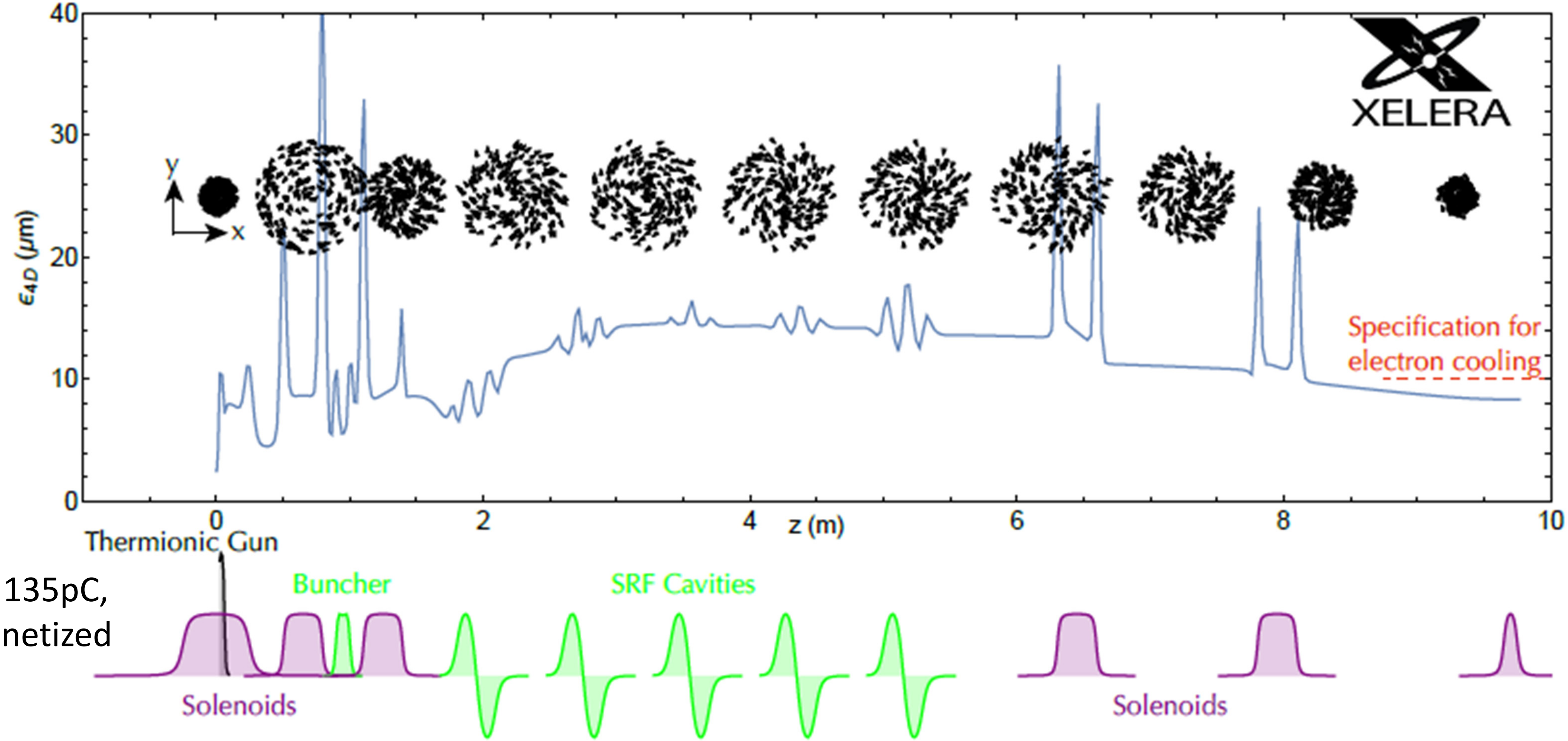
$$I_b = \frac{g_{21}U_{\mathrm{rf}}}{\pi}(\sin\psi - \psi\cos\psi)\,.$$

$$\sigma_t \approx \psi/\omega\sqrt{5}$$

$$\sigma_t = \left(\frac{1}{q_b}\int_{-\psi/\omega}^{+\psi/\omega} t^2 I(t)dt\right)^{1/2} = \frac{1}{\omega\sqrt{3}}\left(\frac{2\psi^3\cos\psi}{\sin\psi - \psi\cos\psi} + 3\psi^2 - 6\right)^{1/2}$$


ERL (bottom)
Tgun
XELERA
magnetized gun
beam dump
linac
booster
chirper
chirper
dipole
dipole
CCR (top)
ion beam
cooling solenoids
magnetization flip
magnetization flip
DC & RF kicker
septum
DC & RF kicker
DC & RF kicker
DC & RF kicker
CCR
ERL
chirper
chirper
injected bunch orbit
extracted bunch orbit
dipole
dipole
Beam Exchange (side)

Phase I: Magnetized Injector Design



Shown for 135pC, 36um magnetized emittance